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Supplement

Title: Targeting alcohol use in high-risk population groups: A US microsimulation study of beverage-specific pricing policies

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Supplement 1. ODD protocol	3
1.1 Overview	3
1.1.1 Purpose and patterns	3
1.1.2 Entities, state variables and scales	3
1.1.3 Process overview & scheduling	4
1.2 Design concepts	6
1.2.1 Stochasticity	6
1.2.2 Observation	7
1.3 Details.....	8
1.3.1 Initialisation	8
1.3.2 Input data	9
1.3.3 Sub-models	11
1.3.4 Model assumptions	21
Supplement 2. Race and ethnicity coding in individual data sources	23
Supplement 3. Model precision across simulation runs	24
Supplement 4A. Simulated average alcohol consumption levels in the total population (main analysis, standard policy setting).....	27
Supplement 4B. Simulated average alcohol consumption levels in the total population (sensitivity analysis 1, more conservative policy setting).	29
Supplement 4C. Simulated average alcohol consumption levels in the total population (sensitivity analysis 2, less conservative policy setting).	31
Supplement 5A. Simulated prevalence of alcohol use categories (main analysis, standard policy setting).....	33
Supplement 5B. Simulated prevalence of alcohol use categories (sensitivity analysis 1, more conservative policy setting).	35
Supplement 5C. Simulated prevalence of alcohol use categories (sensitivity analysis 2, less conservative policy setting).	37
Supplement 6A. Simulated average consumption levels by alcohol use category as of 2018 (main analysis, standard policy setting).....	39
Supplement 6B. Simulated average consumption levels by alcohol use category as of 2018 (sensitivity analysis 1, more conservative policy setting).	43

Supplement 6C. Simulated average consumption levels by alcohol use category as of 2018 (sensitivity analysis 2, less conservative policy setting).	47
Supplement 7. Differences in change in simulated average alcohol consumption by alcohol use category between reference (no price change) and scenario 4: 50% increase for beer and spirits and 10% increase for wine within population subgroups.....	51
References	52

Supplement 1. ODD protocol

1.1 Overview

1.1.1 Purpose and patterns

The purpose of the microsimulation is to estimate the effects of alcohol pricing policies on alcohol consumption levels within population subgroups. A description of the overall study design for the Simulation of Alcohol Control Policies (SIMAH) project can be found in Probst et al. 2023.¹ In the following, we provide full details using the Overview, Design concepts and Details (ODD) protocol for the description of simulation models.²

1.1.2 Entities, state variables and scales

The entities in the model are synthetic individuals that are representative of the (entire) adult US population. The model is designed to be representative over the time period 2000 to 2019. Individuals in the micro-synthetic population are characterised by static and time varying variables listed in Table S1 including age, sex, race and ethnicity, and educational attainment, as well as information on their alcohol use including drinking status (current drinker, abstainer, former drinker) and average grams of pure alcohol consumed per day (i.e., level of consumption). Each time step of the model represents one year.

Table S1. Properties of synthetic individuals.

Parameter	Values	Data source(s)	Update rules
Age	18-79 (individual years)	The age, sex, race and ethnicity and education distributions were assigned based on the Census and American Community Survey (2000-2019) ³	Age increases by +1 year at each time step
Sex	Male, Female		Static properties
Race and ethnicity	non-Hispanic White, non-Hispanic Black, Hispanic, Other		
Educational attainment	High school degree or less, Some college, College degree or more		Individuals aged 18-34 years have the opportunity to update their educational attainment level at each time step. Markov model: transition probabilities conditional on age, sex, and race and ethnicity (described in 1.3.3 Sub-models) ⁴
Drinking status	Current drinker,	2005 National Alcohol Survey data on lifetime abstainer, past year abstainer, past	Current drinker and abstainer are assigned based on alcohol

Parameter	Values	Data source(s)	Update rules
	Former drinker, Abstainer	30-day abstainer and current drinker were used to adjust past 30-day drinking from annual Behavioural Risk Factor Surveillance System (BRFSS) data (2000 to 2019) to past 12 months drinking and to add the category of former drinkers, using distributions from subgroups based on age, sex, race and ethnicity. Further details are described in a published report. ⁵	updating functions. Former drinker is assigned separately by sampling from within abstainer category.
Alcohol use category	Non-drinker, Category 1, Category 2, Category 3	BRFSS 2000 to 2019 data on quantity and frequency of alcohol use in past 30 days were used to calculate average grams of pure alcohol consumed per day (g per day). National- and state-level alcohol per capita consumption data were used to correct for underreporting of alcohol use in survey data, see ⁵ for further details. Discrete alcohol use categories are non-drinker, category 1 with up to 20 g per day for females and up to 40 g per day for males, category 2 with more than 20 g per day up to 40 g per day for females and more than 40 g per day up to 60 g per day for males, and category 3 with more than 40 g per day for females and more than 60 g per day for males. ⁶	All individuals have the opportunity to update at each time step. This is based on an ordinal logistic regression conditional on age, sex, race and ethnicity, educational attainment, and previous drinking (see 1.3.3 Sub-models). For our preliminary work on alcohol transitions, see ⁷ .
Alcohol consumption level	Average grams of pure alcohol consumed per day, 0-200 (capped)		Re-assigned every year in individuals who changed their alcohol use category based on distributions of alcohol consumption levels by alcohol use category and individual characteristics in the BRFSS data (see 1.3.3 Sub-models).
Beverage-specific alcohol use	Proportion (%) of alcohol consumed by beer, wine, and spirits	National Epidemiologic Survey on Alcohol and Related Conditions (NESARC) wave I (2001-2002) and III (2012-2013) For our preliminary work on beverage-specific alcohol use, see. ⁸	Applied only once. Sampled from the distribution of the proportions of alcohol consumed by beer, wine, and spirits conditional on age, sex, race and ethnicity, education, and alcohol use category (see 1.3.3 Sub-models).

1.1.3 Process overview & scheduling

Figure S1 depicts the process overview. The model proceeds in discrete annual time steps. The micro-synthetic population is initialised in the year 2000. Each year, the following process takes place:

- 1) Record the current micro-synthetic population and its properties, including alcohol consumption (i.e., alcohol use category, alcohol consumption level);

- 2) Apply death counts, i.e., remove random sample of individuals in proportion to all-cause mortality rates by age, sex, race and ethnicity, educational attainment informed by National Vital Statistics System (NVSS) data for that year (2000 to 2019);
- 3) Update educational attainment conditional on age, sex, race and ethnicity, this happens for all individuals aged 18-34 and is controlled through a Markov Model;
- 4) Update alcohol use category conditional on age, sex, race and ethnicity, and education, this happens for all individuals and is controlled through an ordinal logistic regression;
- 5) If individuals have changed their alcohol use category, update alcohol consumption level (g per day) based on empirical distributions estimated from the Behavioral Risk Factor Surveillance System (BRFSS) data conditional on alcohol use category and age, sex, race and ethnicity, and educational attainment;
- 6) Age all individuals by +1 year, remove all individuals that are older than 79 years of age; and
- 7) Add and remove individuals due to birth (18-year-olds) and migration based on BRFSS data.

In the year of the policy, the anticipated policy effect is applied before recording alcohol consumption, so that the policy effect is measurable within the same year (i.e., immediate policy effect). The policy modelling process comprises four steps:

- a) Assign beverage-specific alcohol consumption levels for all individuals who currently drink alcohol, based on their empirical distribution based on National Epidemiologic Survey on Alcohol and Related Conditions (NESARC) wave I and III data and conditional on alcohol use category and age, sex, race and ethnicity, and educational attainment;
- b) Apply participation elasticity, accounting for each individual's probability to stop drinking alcohol given their alcohol use category based on pseudo-longitudinal BRFSS data, i.e., this probability is extracted from the ordinal logistic regression model used to update the alcohol use category in each year;
- c) Apply beverage-specific consumption elasticity, accounting for each individual's beverage-specific alcohol consumption levels; and
- d) Update alcohol use categories given new continuous alcohol consumption levels.

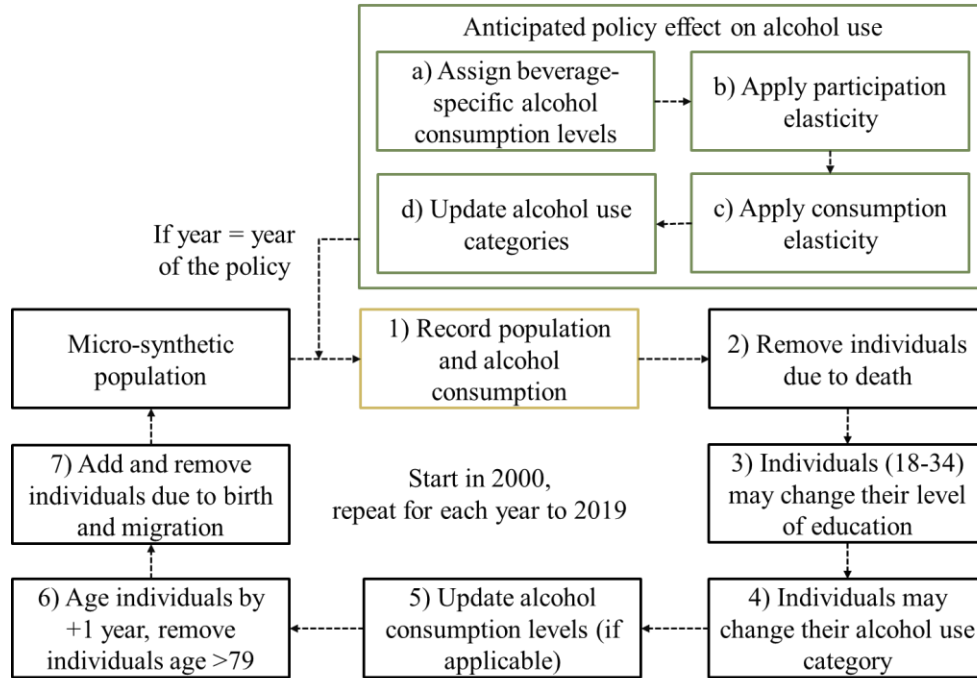


Figure S1. Process overview. Green box: policy process that is applied only if it is the year of the policy; yellow box: recording of population and alcohol consumption within population subgroups (micro-synthetic population remain unchanged).

1.2 Design concepts

1.2.1 Stochasticity

Pseudo-random number (PRN) sampling is used throughout the simulation to assign simulated individuals with characteristics and behaviour. Stochasticity is introduced by varying the seed for the PRN generator. There are different ways in which stochasticity is introduced in the model.

In a first set of mechanisms, individuals have a probability of changing their characteristics or behaviours. A PRN between 0 and 1 is sampled for each individual and their individual probability (e.g., to die) is compared to this number. This applies to:

- *Remove individuals due to death:* Stochasticity is introduced by comparing probabilities of all-cause mortality to a PRN from a uniform distribution at each time step. The probability of dying was generated using each individual's characteristics (i.e., age, sex, race and ethnicity, and educational attainment) and NVSS data (see 1.3.2 Input data). Individuals are assigned as dead if the PRN exceeds the probability, and consequently removed from the simulation.
- *Updating educational attainment:* Education transitions incorporate stochasticity to determine which individuals younger than 35 transition to higher levels of education. The function uses a PRN drawn at each time step to determine the new education level for each individual. The education level is updated based on where the PRN falls within cumulative transition probability thresholds conditional on age, sex, race and ethnicity based on Panel Study of Income Dynamics (PSID) data (see 1.3.2 Input data).
- *Updating alcohol use categories:* Stochasticity is introduced by the ordinal logistic regression model using pseudo-longitudinal BRFSS data assigning individuals a probability

of moving to a different alcohol use category conditional on age, sex, race and ethnicity, and educational attainment. A PRN drawn for each individual at each time step is compared to that probability to decide which categories people may move to.

Stochasticity is also introduced in:

- *Updating alcohol consumption levels:* There is a beta distribution for each subgroup (defined by age, sex, race and ethnicity, educational attainment, and alcohol use category) that is fitted to the BRFSS data (see 1.3.2 Input data). Individuals who change their alcohol use category are re-assigned a new value indicating their consumption level in g per day based on sampling from these beta distributions.
- *Adding and removing individuals due to birth and migration:* There is a migration (18-79 years of age) and birth rate (18-year-olds) for individuals that need to enter the model at each time step that is based on American Community Survey (ACS) data (see 1.3.2 Input data). The number of individuals that need to enter are randomly sampled (by age, sex, race and ethnicity) from each corresponding year of the BRFSS data. There is also an outward migration rate for individuals that need to leave the model in each year by age, sex, race and ethnicity. The number of individuals to leave are randomly sampled from the simulation and removed.
- *Assigning beverage-specific alcohol consumption:* Each individual's beverage-specific alcohol consumption is sampled from their beverage-specific distribution conditional on age, sex, race and ethnicity, educational attainment, and alcohol use category using data from NESARC wave I and III (see 1.3.2 Input data).
- *Applying the policy effect on drinking participation (participation elasticity):* Individuals are randomly sampled given their probability to stop drinking alcohol by alcohol use category. Their probability to stop drinking is calculated from the ordinal logistic regression model used to update alcohol use categories at each time step using pseudo-longitudinal BRFSS data.
- *Estimating an individual-level policy effect (consumption elasticity):* Individual-level consumption elasticities are sampled from the beverage-specific distributions of own-price elasticity accounting for a U-shaped relationship between individual-level consumption elasticity and alcohol consumption level (for details, see 1.3.3 Sub-models).⁹ A medium correlation of $r = .60$ is applied in the main model, while alternative correlation coefficients are used in sensitivity analyses (see manuscript).

1.2.2 Observation

The observations are separated into (1) within the simulation loop, observed at each time step (i.e., year), and (2) the results outside the simulation loop, summarised post simulation.

For (1): Population data is stored and updated at each time step in the *basepop* variable. This contains basic general information on each individual; a unique id, age, sex, race and ethnicity, educational attainment, binary drinking status, alcohol consumption level (in g per day), alcohol use category, simulation entry year, and age in categorical form. Alcohol consumption data is stored at each time step and as discrete alcohol use categories and consumption levels of all individuals, alongside sociodemographic data (i.e., age, sex, race and ethnicity, and educational attainment).

For (2): Summary data, depending on the output specified, is processed and provided at the end of the simulation. Alcohol can be specified as the observed output, which provides stratified

information of alcohol consumption at each time step. Three different output formats are available, providing alcohol consumption data by sex and educational attainment (can also include age and race and ethnicity as stratifying variables): *alcoholcont* provides mean alcohol consumption levels, *alcoholcat* gives the proportion of individuals within each alcohol use category, and *alcoholcontcat* provides mean alcohol consumption levels by alcohol use category as of the year before the policy. For all output formats, the summary data also provides 95% confidence intervals (CI) capturing the inter-individual heterogeneity within each subgroup and time step.

1.3 Details

1.3.1 Initialisation

Individuals are initialised to form a representative population of the resident US adult population (18 to 79 years of age). Default population size is set to 1,000,000 individuals but can be altered, generally by powers of 10.

There are three stages of model initialisation:

- 1) Retrieve population constraints (R script: *1_generate_IPF_constraints*): This is the raw data processing of the population data provided by the ACS, and corresponds to the US census in decennial years (2000, 2010). It removes all individuals older than 79 and converts variables for age categories, sex, race and ethnicity, and educational attainment into new categorical values. It then groups the data by these variables and an encoded state identification number to extract the population weights of each sociodemographic group. The script outputs a summary CSV-file, showing the weights of each sociodemographic group at the national level, with both numerical and name identifiers. These weights are used as the population constraints.
- 2) Generate weights for iterative proportional fitting¹⁰ (IPF; R script: *2_Calculate_population_weights*): Data from the population constraints generated and the BRFSS to capture alcohol consumption are used. The data is processed to select the required variables in the specified year and ensure the data is ready for IPF (see *microsimpackage::process_for_IPF*). The weights are then generated for the dataset (*microsimpackage::generate_weights_IPF*). Within this, it filters the population constraints and the BRFSS data to include only the regional level (US), creates a binary matrix of demographic categories, and aligns the individual data with the population constraints. The IPF algorithm (*ipfp*) adjusts initial weights to ensure the weighted individual data matches the population constraints. Finally, it converts these weights to integers and returns a data frame with individual IDs, and their corresponding weights.
- 3) Generate the baseline population (R script: *3_generate_population*): The synthetic baseline population is generated using the weights calculated above. The weights are initially normalised, scaled, and rounded to the nearest integer to match the population size. The synthetic population is created by expanding the population data according to the new weights, and then sampling n rows of the expanded population, where n is the desired population size, to create a representative synthetic population. To ensure that the data is in the correct format for future use, the data is post processed. It creates a flag for former drinkers, standardising sex coding, selecting and renaming columns for microsimulation compatibility, and adding a spawn year, and age categories. It then saves the processed data to a CSV-file named according to the selected region (US) and population size.

1.3.2 Input data

The population size in each cross-tabulated sociodemographic category (e.g., Black, non-Hispanic males aged 18-24 with low education) is determined using data from the ACS and US census, which provide the share of individuals in each category. Individuals with specific attributes, including alcohol consumption, are then sampled based on the BRFSS data to create the synthetic baseline population with the default population size of a million individuals. An overview of all individual-level data sources used to inform the microsimulation is provided in Table S2.

Table S2. Overview of individual-level data sources used in the microsimulation.

Data source	Years	Design	Sampling	Purpose
US Census	2000, 2010	Cross-sectional	Full assessment of the US population	To obtain population counts by population subgroup for baseline population and modelling birth and migration, downloaded from IPUMS ³
American Community Survey (ACS)	Annual, 2000-2019	Cross-sectional	US civilian population, representative on the national and the state level; institutionalised populations and people living in grouped quarters included since 2006	
Panel Study of Income Dynamics (PSID)	Biennial, 1999-2019	Cohort	US civilian, non-institutionalised population, nationally representative	To inform prior beliefs for the educational attainment transition model
Behavioral Risk Factor Surveillance System (BRFSS)	Annual, 2000-2019	Cross-sectional	US civilian, non-institutionalised population, representative on the national and the state level	To derive individual's characteristics and to inform prior beliefs for alcohol consumption transition model; data corrected for underreporting of alcohol use in survey studies (see ⁵)
National Epidemiologic Survey on Alcohol and Related Conditions (NESARC)	2001-2002 (NESARC wave I), 2012-2013 (NESARC wave III)	Cross-sectional	US civilian, non-institutionalised population; nationally representative	To obtain distribution of beverage-specific alcohol consumption by population subgroup and alcohol use category
National Vital Statistics System (NVSS)	Annual, 2000-2019	Registry	Full assessment of individual death records	To inform mortality rates

IPUMS: Integrated Public Use Microdata Series.³ All data sources are restricted to individuals aged 18 years and older.

1.3.3 Sub-models

Apply death counts

Mortality is modelled using death counts in each population subgroup, taken directly from the 2000-2019 NVSS data and scaled to the population size that is being modelled, calculating a corresponding mortality rate. The current sub-model accounts for all-cause mortality.

Updating educational attainment

The microsimulation used data from the US Census to form the baseline estimates and data from PSID to update individual's educational attainment categories over time. In the US Census data, educational attainment is reported in terms of grades of school completed in the following categories: (1) Less than 9th grade, (2) 9th to 12th grade, no diploma, (3) High school graduate, (4) Some college, no degree, (5) Associate degree, (6) Bachelor's degree and (7) Graduate or professional degree. This data was used to inform the education distribution in the microsimulation at baseline, where category (3) and below represents "High school degree or less", categories (4) and (5) will be classified as "Some college" and categories (6) and above will be classified as "College degree or more". In the American Community Survey data, educational attainment is reported in terms of years of education completed, and ranges from no schooling, through to grade 4-12 and then 1-5+ years of college. We categorized ACS data into the following categories: Grade 12 and below will be classified as "High school degree or less", 1-3 years of college will be classified as "Some college" and 4 and 5+ years of college will be categorized as "College degree or more". In the PSID data, educational attainment is also classified in terms of years of education completed and possible values range from 1-17 years. We classified 12 years or less as "High school degree or less" and 13-15 years as "Some college" (representing, 1, 2 and 3 years of college, respectively). Education of 16 years or over was considered as "College degree or more".

Education transitions are modelled using multi-state Markov models with a homogeneous, continuous-time process¹¹ and 2005-2019 longitudinal PSID data, restricting changes in educational attainment to individuals aged 18 to 34 years.⁴ Educational attainment is therefore assumed to be fixed after age 34, as post-graduate degrees are assumed to have little impact on alcohol use.

The model estimates transitions between 5 states: (1) High school degree or less, (2) Some college (1 year), (3) Some college (2 years), (4) Some college (3 years), (5) College degree or more. The model is estimated conditional on age category (18, 19, 20, 21 as individual years of age, 22-24, 25-29, 30+ as age categories), sex, and race and ethnicity. Allowed instantaneous transitions are restricted to sequential forward transitions to the next higher level of education (see Q matrix below). We estimate three separate years in the "some college" category as tunnel states, which prevents individuals from achieving higher educational attainment categories too early (e.g., this would prevent individuals from reaching the "College degree or more" category in 1 year). There is no time component to the model, i.e., education transition probabilities are the same throughout the simulation 2000 to 2019.

To calculate the transition probabilities for a specific sociodemographic subgroup, we start with the baseline transition intensities (Table S3) and adjust these values using Hazard Ratios (HRs) for the relevant sociodemographic properties of that subgroup (Table S4). The allowed transitions are defined in the Q matrix:

$$\begin{pmatrix} -q_{12} & q_{12} & 0 & 0 & 0 \\ 0 & -q_{21} + q_{23} & q_{23} & 0 & 0 \\ 0 & 0 & -q_{32} + q_{34} & q_{34} & 0 \\ 0 & 0 & 0 & -q_{43} & q_{43} \\ 0 & 0 & 0 & 0 & -q_{54} \end{pmatrix}$$

The baseline transition intensity matrix (Q) defines the instantaneous rates of transitioning between states under baseline conditions (Table S3)—i.e., the values of the covariates are at the reference values. Each element q_{ij} represents the baseline intensity of transitioning from state i to state j and the diagonal elements q_{ii} are calculated to ensure that the rows sum to zero.

$$q_{ii} = - \sum_{j \neq i} q_{ij}$$

Table S3. Baseline (top) and calibrated (bottom) transition intensities (before adjusting for covariates).

State T	State T+1 High school degree or less	Some college (1 year)	Some college (2 years)	Some college (3 years)	College degree or more
			Intensity (95% CI)		
High school degree or less	.	0.23 (0.21,0.25)	.	.	.
Some college (1 year)	.	0.29 (0.28,0.29)	1.00 (0.90,1.10)	.	.
Some college (2 years)	.	.	1.01 (0.99,1.01)	0.57 (0.53,0.61)	.
Some college (3 years)	.	.	.	0.51 (0.50,0.51)	1.24 (1.18,1.30)
College degree or more	.	.	.	.	1.10 (1.08,1.11)

Using the values from Table S3, the baseline Q matrix is:

$$Q = \begin{pmatrix} -0.23 & 0.23 & 0 & 0 & 0 \\ 0 & -1.00 & 1.00 & 0 & 0 \\ 0 & 0 & -0.57 & 0.57 & 0 \\ 0 & 0 & 0 & -1.24 & 1.24 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

The baseline intensities are adjusted for subgroup specific characteristics using the HRs from Table S4. For a given transition $i \rightarrow j$, the adjusted transition intensity λ_{ij} is calculated as:

$$\lambda_{ij} = q_{ij} \cdot HR_{ij}$$

For example, for a white, male, 19-year-old transitioning between states $\leq$ High school (high school degree or less) $\rightarrow$ SomeC1 (some college, 1 year):

$$\lambda_{\leq High\ school \rightarrow SomeC1} = 0.23 \cdot 0.72 = 0.1656$$

The adjusted Q_{group} matrix for a 19-year-old White male is therefore:

$$Q_{\text{group}} = \begin{pmatrix} -0.1656 & 0.1656 & 0 & 0 & 0 \\ 0 & -0.77 & 0.77 & 0 & 0 \\ 0 & 0 & -0.609 & 0.609 & 0 \\ 0 & 0 & 0 & -0.8184 & 0.8184 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

The transition probability matrix $P(t)$ is calculated using the matrix exponential of the adjusted intensity matrix:

$$P(t) = \exp(Q_{\text{group}} \cdot t)$$

This must be calculated using statistical software such as R; the full transition probabilities used in our model are available online: <https://figshare.com/s/f21a5d3a01258d5007bb>; DOI:10.6084/m9.figshare.28574336.

Table S4. Hazard ratios (HR) for each sociodemographic factors' influence on the transition intensities, calibrated model version.

		Original model (estimated using PSID data)				Calibrated model (calibrated to ACS 2000-2010, validated to ACS 2011-2019)			
		≤High school → SomeC1	SomeC1 → SomeC2	SomeC2 → SomeC3	SomeC3 → ≥College	≤High school → SomeC1	SomeC1 → SomeC2	SomeC2 → SomeC3	SomeC3 → ≥College
Age	18	1	1	1	1	1	1	1	1
	19	0.62 (0.61,0.62)	0.78 (0.77,0.79)	1.27 (1.25,1.28)	0.74 (0.73,0.75)	0.72 (0.72,0.72)	0.77 (0.76,0.78)	1.07 (1.06,1.08)	0.66 (0.66,0.67)
	20	0.23 (0.23,0.23)	0.5 (0.50,0.51)	1.01 (1.00,1.02)	0.73 (0.72,0.75)	0.28 (0.28,0.29)	0.54 (0.53,0.54)	0.92 (0.91,0.93)	0.73 (0.73,0.74)
	21	0.16 (0.16,0.17)	0.32 (0.31,0.32)	0.65 (0.64,0.66)	0.59 (0.58,0.6)	0.24 (0.24,0.25)	0.33 (0.33,0.33)	0.62 (0.62,0.62)	0.43 (0.42,0.43)
	22-24	0.12 (0.12,0.12)	0.19 (0.19,0.20)	0.31 (0.31,0.31)	0.39 (0.38,0.39)	0.12 (0.12,0.13)	0.19 (0.19,0.19)	0.37 (0.37,0.38)	0.33 (0.32,0.33)
	25-29	0.11 (0.11,0.12)	0.18 (0.18,0.18)	0.19 (0.19,0.19)	0.2 (0.2,0.2)	0.12 (0.12,0.12)	0.21 (0.21,0.21)	0.19 (0.19,0.19)	0.18 (0.18,0.18)
	30-34	0.1 (0.1,0.11)	0.14 (0.14,0.14)	0.15 (0.15,0.15)	0.22 (0.21,0.22)	0.09 (0.09,0.09)	0.13 (0.13,0.14)	0.19 (0.18,0.19)	0.19 (0.19,0.19)
Sex	Males	1	1	1	1	1	1	1	1
	Females	1.34 (1.33,1.34)	1.18 (1.18,1.19)	1.07 (1.07,1.08)	1.14 (1.13,1.14)	1.42 (1.42,1.42)	1.25 (1.23,1.26)	1.09 (1.08,1.11)	1.06 (1.05,1.07)
Race and ethnicity	Non-Hispanic White	1	1	1	1	1	1	1	1
	Non-Hispanic Black	0.71 (0.7,0.72)	0.73 (0.73,0.74)	0.85 (0.84,0.86)	0.58 (0.57,0.58)	0.69 (0.68,0.70)	0.76 (0.76,0.77)	0.73 (0.72,0.75)	0.64 (0.64,0.65)
	Other	0.89 (0.88,0.90)	0.76 (0.75,0.77)	0.89 (0.88,0.9)	0.54 (0.54,0.55)	0.79 (0.79,0.79)	0.9 (0.90,0.91)	0.88 (0.87,0.89)	0.63 (0.63,0.64)
	Hispanic	1.19 (1.18,1.21)	1.04 (1.03,1.05)	1.16 (1.15,1.18)	0.82 (0.81,0.84)	1.14 (1.12,1.16)	1.02 (1.01,1.03)	1.27 (1.26,1.28)	0.73 (0.73,0.74)

All values are Hazard Ratios (HRs), 95% confidence intervals in brackets. ACS: American Community Survey, ≤High school: High school degree or less, SomeC1: some college 1 year, SomeC2: some college 2 years, SomeC3: some college 3 years, ≥College: College degree or more.

Calibration and validation

Education transition probabilities are calibrated to match the observed proportions of individuals in each educational attainment category by age, sex, and race and ethnicity over time according to the observed target data using 2000-2010 and validated using reserved data from 2011-2019 ACS.¹² The education transition probabilities were a good fit to the data and the calibration process converged after 10 waves of calibration with a mean implausibility metric of 2.21 across all samples. All calibrated samples passed the validation criteria (mean implausibility: 2.70) for the validation period 2011-2019. The implausibility metric is calculated using three key sources of uncertainty: observation uncertainty, represented by the measurement error of the observed targets; ensemble variability, quantified as the 90th percentile of the sample variance; and model discrepancy, measured as 10% of the variance of the simulator outputs.¹³

Updating alcohol consumption

Transitions between alcohol use categories (i.e., non-drinker, category 1 (low risk, up to 20 g per day for females and up to 40 g per day for males), category 2 (medium risk, more than 20 g per day up to 40 g per day for females and more than 40 g per day up to 60 g per day for males), category 3 (high risk, more than 40 g per day for females and more than 60 g per day for males) conditional on previous alcohol use category, age category (i.e., 18-24, 25-64, 65+), sex, race and ethnicity, and educational attainment were obtained from ordinal logistic regression. The regression model was fitted to “deterministic” data that has been generated using the simulation and the 2000-2010 BRFSS data. This approach has been chosen due to issues fitting the simulation to the observed data (BRFSS 2015-2019) using continuous-time Markov models, as proposed in the study protocol.^{1,7}

The “deterministic” population was generated by using a function to impute daily alcohol consumption levels (in g per day) for each synthetic individual based on the BRFSS data. The function ranks synthetic individuals (within age, sex, race and ethnicity, and education categories) and matches them with a BRFSS individual by rank, assigning alcohol values accordingly. This results in a “deterministic” population that mirrors BRFSS alcohol consumption data for the synthetic population, providing a longitudinal dataset for ordinal logistic regression model estimation (see Figure S2).

An ordinal logistic regression model was fitted to the pseudo-longitudinal BRFSS data (i.e., the deterministic population) using alcohol use category in year t as outcome variable and age, sex, race and ethnicity, and educational attainment in year $t - 1$ as predictors. Interaction effects between age X alcohol use category, sex X age, sex X race and ethnicity, sex X educational attainment, and educational attainment X alcohol use category were included.

The fitted ordinal logistic regression is used to calculate a probability of each individual belonging to each alcohol use category, conditional on age, sex, race and ethnicity, and educational attainment. A PRN is sampled for each individual and compared to these probabilities to assign each individual a new alcohol use category in each year of the simulation. In the ordinal logistic regression model, the non-drinker category includes both former drinkers and lifetime abstainers. Each year within the simulation, a random subset of non-drinkers is selected and reclassified as former drinkers based on the proportion of former drinkers observed in the BRFSS data. The remaining non-drinkers are classified as lifetime abstainers.

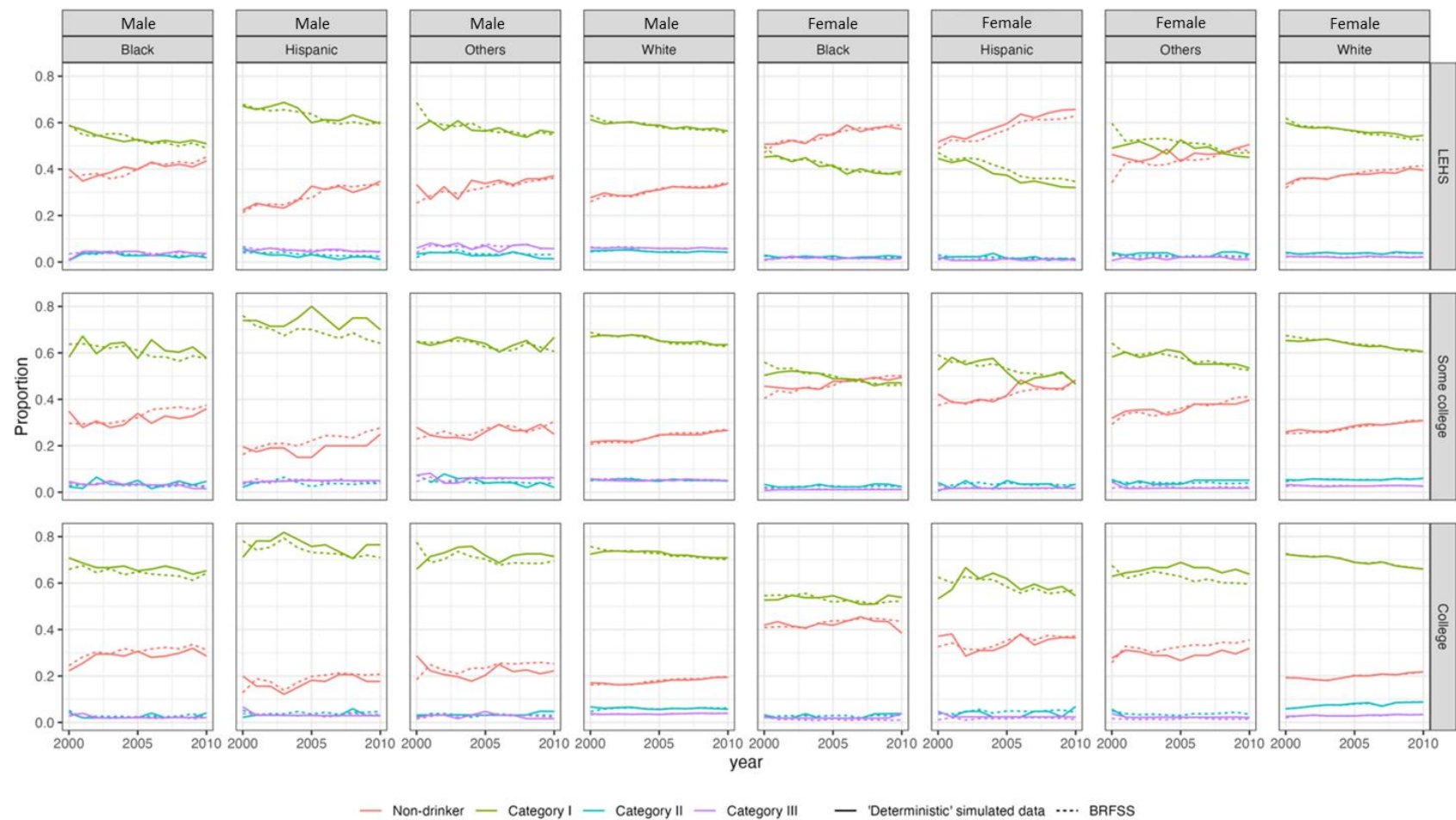


Figure S2. Proportion of individuals in each alcohol use category by sex, race and ethnicity and educational attainment in Behavioral Risk Factor Surveillance System (BRFSS) data compared to 'deterministic' pseudo-longitudinal BRFSS data used to fit ordinal logistic regression models. LEHS: high school degree or less, College: college degree or more.

Calibration and validation

The ordinal logistic regression model was calibrated to observed target data from the BRFSS (2011-2015) and were of good fit. The calibration process converged after 4 waves of calibration with a mean implausibility metric of 2.39 across all samples. All calibrated samples passed the validation criteria (mean implausibility: 2.41) for the period 2016-2019. Table S5 depicts the full specification of the ordinal logistic regression model before and after calibration.

Table S5. Original and calibrated Ordinal Logistic Regression model for modelling alcohol use transitions.

	Original model (estimated using pseudo-longitudinal BRFSS data)			Calibrated model (calibrated to BRFSS 2011-2015, validated to BRFSS 2016-2019)		
	OR	Lower	Upper	OR	Lower	Upper
Alcohol use category 1 ^a	26.25	23.04	29.89	41.78	41.78	41.78
Alcohol use category 2 ^a	469.46	376.70	585.07	65480.22	65480.17	65480.27
Alcohol use category 3 ^a	1789.76	1381.30	2319.00	4002351.10	4002346.69	4002355.52
Females ^b	1.02	0.91	1.14	0.96	0.96	0.96
Age 25-64 ^c	0.89	0.80	0.99	0.61	0.61	0.61
Age 65+ ^c	0.67	0.59	0.75	0.49	0.49	0.49
≤High school ^d	0.80	0.73	0.86	0.51	0.51	0.51
Some college ^d	1.03	0.94	1.12	0.80	0.80	0.80
Non-Hispanic Black ^e	0.75	0.68	0.81	0.77	0.77	0.77
Hispanic ^e	0.95	0.86	1.04	0.94	0.94	0.94
Non-Hispanic Others ^e	0.93	0.85	1.01	0.96	0.96	0.96
Females ^b * Low education ^d	0.89	0.82	0.96	0.92	0.92	0.92
Females ^b * Medium education ^d	0.90	0.82	0.97	0.94	0.94	0.94
Females ^b * Non-Hispanic Black ^e	0.80	0.72	0.90	0.81	0.81	0.81
Females ^b * Hispanic ^e	0.64	0.57	0.73	0.65	0.65	0.65
Females ^b * Non-Hispanic Others ^e	0.84	0.74	0.95	0.82	0.82	0.82
Females ^b * Age 25-64 ^c	0.99	0.90	1.10	1.06	1.06	1.06
Females ^b * Age 65+ ^c	0.80	0.71	0.90	0.83	0.83	0.83
Category 1 ^a * Age 25-64 ^c	0.69	0.62	0.77	0.92	0.92	0.92
Category 2 ^a * Age 65+ ^c	0.64	0.56	0.73	0.75	0.75	0.75
Category 2 ^a * Age 25-64 ^c	0.94	0.78	1.14	0.80	0.80	0.80
Category 2 ^a * Age 65+ ^c	1.49	1.18	1.88	1.03	1.03	1.03
Category 3 ^a * Age 25-64 ^c	1.22	0.99	1.51	4.44	4.44	4.44
Category 3 ^a * Age 65+ ^c	1.58	1.19	2.10	2.87	2.87	2.87
Category 1 ^a * ≤High school	0.82	0.75	0.89	1.70	1.70	1.70
Category 1 ^a * Some college	0.76	0.69	0.83	1.10	1.10	1.10
Category 2 ^a * ≤High school	1.63	1.41	1.88	1.80	1.80	1.80
Category 2 ^a * Some college	1.21	1.05	1.41	1.25	1.25	1.25
Category 3 ^a * ≤High school	2.04	1.69	2.45	2.97	2.97	2.97
Category 3 ^a * Some college	1.26	1.03	1.53	1.18	1.18	1.18
Non-drinker Category 1	1.63	1.45	1.84	1.69	1.69	1.69
Category 1 Category 2	282.70	248.47	321.64	2758.12	2758.12	2758.12
Category 2 Category 3	1529.46	1334.32	1753.14	627726.59	627726.23	627726.94

BRFSS: Behavioral Risk Factor Surveillance System, ≤High school: High school degree or less, ≥College: College degree or more. All dynamic variables (i.e., age, educational attainment, alcohol use category) use previous year's values ($t-1$) to predict the current year (t). ^a ref = non-drinker, ^b ref = age 18-24, ^c ref = males, ^d ref = ≥College, ^e ref = non-Hispanic White

A separate process is used to assign individuals a continuous g per day value from within their alcohol use category. Individuals are only assigned a new continuous g per day value when they change their alcohol use category. This is controlled by beta distributions fitted to the 2000-2014 BRFSS data for each alcohol use category, age (18-24, 25-64, 65+), sex, race and ethnicity, and educational attainment category. For the 18-24 age group, individuals in the college and some college groups were combined into ‘some college’, as there were too few college graduates within this age range to estimate a separate distribution. This resulted in a total of 92 distinct groups for which beta distributions were fitted.

Alcohol consumption values are scaled to fall within the range 0 to 1 using the below transformation, where GPD = g per day, GPD_{min} = minimum g per day for the alcohol use category and GPD_{max} = maximum g per day for the alcohol use category. The small constants 10^{-10} and 10^{-9} are included to ensure numerical stability.

$$scaledGPD = \frac{(GPD - GPD_{min}) + 10^{-10}}{(GPD_{max} - GPD_{min}) + 10^{-9}}$$

In the microsimulation, individuals are allocated sampled values from the beta distributions corresponding to their subgroup. Values are initially sampled between 0 and 1 and rescaled to reflect the actual range of alcohol consumption (in g per day) for the relevant alcohol use category. The transformation used to rescale the values, where *sampledvalue* is the original value sampled between 0 and 1 is:

$$sampledGPD = ((GPD_{max} - GPD_{min} + 10^{-10}) \cdot sampledvalue) + (GPD_{min} - 10^{-9})$$

To assess how well the beta distributions fits the data, the mean alcohol consumption values from the BRFSS (2000-2019) are compared with those produced by the simulation for each group to see if there are any large differences using the above-describe mean implausibility threshold of 3. This comparison indicated that adjustments were needed only for the group ‘Alcohol use category 1 males aged 65 and over’ across all race and ethnicity groups and within the high school degree or less and some college education groups. For these six groups, the shape and scale parameters of the beta distributions were refined by resampling and rerunning the model to identify parameters that best aligned with the target data. The full specification of beta distributions, including original values derived from BRFSS data and adjusted values for the refined groups is available online: <https://figshare.com/s/f21a5d3a01258d5007bb>; DOI:10.6084/m9.figshare.28574336.

Births and migration

Individuals aged 18 are added to the model in each year based on 2000-2019 ACS data. This is based on the number of 18-year-olds that are in the population in each corresponding year, subtracting the number of 18-year-olds that reported to have migrated into the US within the last year (based on ACS data), as migration was modelled separately.

Inward and outward migrations rates are specified for each year. As no data on outward migration is available (i.e., the US does not routinely collect data on how many residents leave the country), migration rates are estimated through simulation in a pre-modelling step that adjusts the population counts to match those in the 2000-2019 ACS data and calculates the

corresponding migration rates that would be required to achieve these counts. Migration rates are specific to 18-year-olds and 5 years age categories, sex, and race and ethnicity. Prior beliefs for inward migration are calculated based on 2000-2019 ACS data and survey questions that asked individuals whether their place of residence was different one year prior to the survey. Outward migration rates are estimated as part of the pre-modelling simulation given the lack of data.

Applying the policy effect

In the year specified as the year of the policy, all current alcohol users change their alcohol consumption based on beverage-nonspecific own-price participation elasticity (i.e., the share of individuals who stop drinking alcohol given a specific price change)¹⁴ and beverage-specific own-price consumption elasticities (i.e., the proportional change in each individual's consumption level given a specific price change)¹⁵ from the literature. Table S6 gives an overview of the policy effect estimates implemented in the model.

Table S6. Policy effect estimates used in the microsimulation model.

Policy effect	Policy effect estimates ^a	Scope
Own-price participation elasticity ¹⁴	M = -0.297, SE = 0.399 (beverage-nonspecific) ^b	US-specific (estimate based on 2004 beer prices (Uniform Product Code scanner data from AC Nielsen) and beer-specific alcohol use (NESARC wave II) for 25 US states
Own-price consumption elasticity ¹⁵	• Beer: M = -0.52, SE = 0.08^c • Wine: M = -0.55, SE = 0.08^c • Spirits: M = -0.60, SE = 0.08^c	US average based on 36 (beer), 31 (wine), and 40 (spirits) studies covering the period from 1934 to 2000

NESARC: National Epidemiologic Survey on Alcohol and Related Conditions (NESARC).

^a Own-price elasticity can be interpreted as the percentage change in alcohol consumption given a 1% price increase, i.e., a 1% increase in beer prices would result in an average decrease in beer consumption by -0.52%.

^b Standard error was not implemented (see Supplement 3). ^c Standard errors (SE) were calculated based on reported standard deviation and number of studies included in summary measure.

Changes in alcohol consumption levels (g per day) are specific to beverage type (beer [including coolers], wine, spirits) and require the modelling of beverage-specific alcohol use first. Individual-level beverage-specific alcohol use was sampled from the distributions of beverage-specific alcohol use, defined by mean proportions and standard deviations, by age category (18-24, 25-49, 50+), sex, race and ethnicity, educational attainment, and alcohol use category using NESARC wave I and III data (see also ⁸). Calculations accounted for the NESARC sampling design and applied survey weights. We combined NESARC wave I and III to increase the sample size to $n = 51,682$ current alcohol users. We assumed a minimum sample size per group ($k = 216$ groups) of $n = 10$ to avoid that mean proportions were based on too few survey respondents, increasing the risk of imprecise estimates. For groups with $n < 10$ ($k = 35$ groups), we combined alcohol use category 2 and 3, resulting in additional 19 groups with sufficient sample size. For the remaining $k = 16$ groups, we used the beverage preference of the medium age category (25-49) within the same population subgroup defined by sex, race and ethnicity, educational attainment, and alcohol use category.

Next, the participation elasticity (i.e., the proportion of individuals who stop drinking alcohol use given a certain price increase) is applied. From all current alcohol users, we randomly select the number of individuals to stop drinking alcohol according to the beverage-nonspecific own-price participation elasticity (i.e., for a 10% price increase, we expect 2.97% of current alcohol users to stop drinking alcohol), while accounting for individuals of different alcohol use categories to have different probabilities to stop drinking. These probabilities were calculated based on the ordinal logistic regression used for modelling alcohol transitions, resulting in a substantially higher likelihood of category 1 alcohol users to stop drinking alcohol compared to category 2 or 3 alcohol users. As the own-price participation elasticity is generic (i.e., beverage-nonspecific), we use the weighted mean price change for each policy scenario, where beverage-specific price changes were applied. Weights are based on the level of beverage-specific alcohol consumption by alcohol use category. In other words, weights are higher for alcoholic beverages with higher alcohol consumption levels within a specific alcohol use category.

Last, beverage-specific own-price consumption elasticities are applied to individual-level alcohol consumption levels among those who continue to drink alcohol. Empirical studies suggest that there is substantial inter-individual variability in own-price consumption elasticities as a function of alcohol consumption, i.e., that individuals with very low and very high alcohol use are the least likely to change their consumption following a price change.^{9,16} We account for this inter-individual variability in sampling individual-level consumption elasticities (henceforth “percent reduction in alcohol use”) on the assumption of a U-shaped association between elasticity estimates and alcohol consumption levels at the individual level (see ⁹). For each synthetic individual, we hence sample a beverage-specific percent reduction in alcohol use from the beverage-specific distributions of own-price consumption elasticities defined by their mean and standard error. In each microsimulation run, the beverage-specific mean consumption elasticities are randomly drawn from the beverage-specific population distributions as defined in Table S6. As for the standard error, we use renormalized standard errors (i.e., the standard error of the population was rescaled to reflect the standard error of the sampled mean). This approach assumes that the standard error across different levels of alcohol consumption would be the same. A schematic overview of this modelling procedure is illustrated in Figure S3.

The sampling of the beverage-specific percent reductions in alcohol use account for each individual’s alcohol consumption level, applying a U-shaped association with a medium correlation of $r = 0.6$ in the main set up of the modelling. In sensitivity analyses, we implement higher ($r = 0.8$) and lower ($r = 0.4$) correlation coefficients, leading to a smaller and larger inter-individual variability of beverage-specific own-price consumption elasticities (see Figure S4). Last, individual-level beverage-specific percent reductions in alcohol use are multiplied with the beverage-specific price change of the policy scenario (e.g., 10%) and the individual-level, beverage-specific alcohol consumption level, resulting in new beverage-specific alcohol consumption levels. For each synthetic individual, beverage-specific alcohol consumption is totalled to determine the level of alcohol consumption in g per day after the policy.

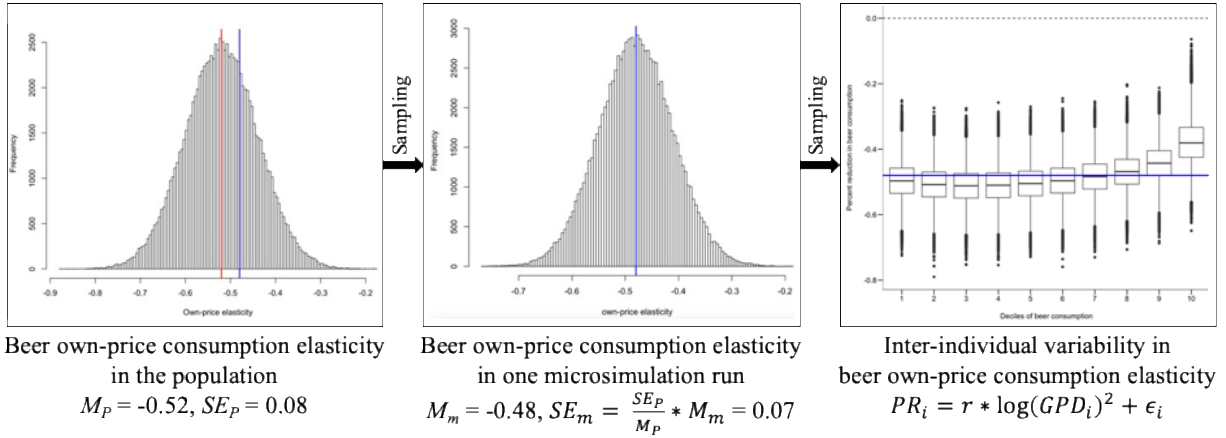


Figure S3. Schematic overview of modelling inter-individual variability in beverage-specific own-price consumption elasticities using the example of beer. The red line indicates mean own-price consumption elasticity of the population defined by its mean (M_P) and standard error (SE_P); blue lines indicate mean own-price consumption elasticity randomly drawn from population distribution and is defined by a new mean (M_m) and the renormalised standard error (SE_m). The individual-level percent reduction in beer consumption (PR_i) is sampled from the own-price consumption elasticity distribution defined by M_m and SE_m as a linear function of the squared logarithm of beer-specific alcohol consumption in g per day (GPD_i), the correlation coefficient (r), and an error term (ϵ_i).

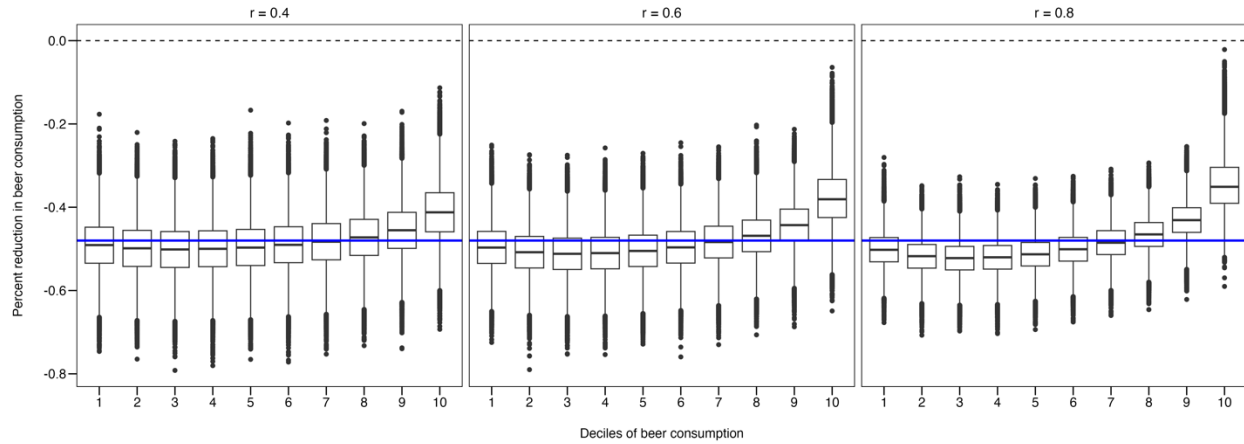


Figure S4. Scatter plot of individual-level percent reduction in beer consumption and deciles of log beer consumption by strength of correlation. Blue lines indicate mean own-price consumption elasticity.

1.3.4 Model assumptions

To emulate and reduce real-world complexity, the microsimulation model is based on three global assumptions: i) the data used to inform synthetic population and model parameters are valid and generalizable to the adult US population and its subgroups across time; ii) the structure of the Markov and ordinal model used to inform transition processes in education and alcohol use, respectively, are reasonable representations of the corresponding population dynamics;^{4,17} iii) the implemented policy mechanisms provide a valid approximation of real-world price policy effects. Within those global assumptions we deem the following as most critical to the presented results: i) beverage preferences informed by two waves of NESARC data are an accurate representation of beverage preferences across US subgroups over time⁸ and upshifting

procedures applied to BRFSS data resulted in a valid and equally accurate representation of drinking behaviours in all population subgroups;⁵ ii) drinking behaviours are relatively stable over time as represented in the pseudo-panel generated to inform the ordinal model of alcohol use transitions; iii) the implemented increases in mean prices are uniform across the population; iii) the mean effects of price increases are solely conditional on beverage category and the current level of alcohol use but independent of the pre-existing price or the sociodemographic subgroup;¹⁸ and iii) price increases in one beverage category do not affect consumption in other beverage categories (i.e., the absence of cross-elasticities). Another key assumption we made with our model was that education is a sufficient indicator of socioeconomic status in the US. All of these assumptions had to be made due to a lack of data or better empirical evidence.

Supplement 2. Race and ethnicity coding in individual data sources

Table S7. Race and ethnicity coding in individual data sources.

Data source	Assessment of Hispanic ethnicity	Imputation of race and ethnicity in this study	Handling of multiple-race groups
American Community Survey (ACS) and US Census	Separate question on Hispanic origin	N/A	Use variable <i>RACE</i> , which classifies into 1. White, 2. Black, 3. American Indian/Alaska Native, 4. Chinese, 5. Japanese, 6. Other Asian or Pacific Islander, 7. Other race, 8. Two major races, 9. Three or more major races. 3-9 are classified as “other” race
Panel Study of Income Dynamics (PSID)	Separate question on Hispanic origin since 2005; an option to report a race that may be considered ‘Hispanic’ was included between 1990 and 2003 (e.g., “Spanish-American”, “Puerto Rican, Mexican”)	A complex imputation method based on hierarchical decision tree (e.g. taking self-report over reported by a spouse) was developed; see Buckley et al. 2025. ⁴	‘First response’ race was considered only, i.e., multiple-race category was not included
National Vital Statistics System (NVSS)	Separate question on Hispanic origin; identical question since 2003, between 2000 and 2002 similar question but with fewer answering options	All deaths with unknown race and/or ethnicity are assigned to “other” race and ethnicity group	Bridged race variable provided (<i>488 Hispanic Origin/Race Recode</i>); single race assigned by data owners using regression models and 1997-2000 census data ¹⁹
Behavioral Risk Factor Surveillance System (BRFSS)	Separate question on Hispanic origin	N/A	No reporting of multiple race
National Epidemiologic Survey on Alcohol and Related Conditions (NESARC)	Separate question on Hispanic origin	N/A	Individuals with multiple races were assigned to “other” race group; for NESARC wave III this was done by the data owners

Supplement 3. Model precision across simulation runs

Two different types of uncertainty are accounted for: 1) uncertainty through stochasticity (see 1.2.1 Stochasticity) and 2) parameter uncertainty (i.e., posterior beliefs of the calibrated education and alcohol models, as well as uncertainty of the policy effect estimate). To identify the number of simulation runs required to obtain robust model estimates in the primary model outcome of interest (i.e., mean alcohol consumption level) for both types of uncertainty, we investigated the impact of running multiple iterations of the microsimulation on model precision. Our aim was to eliminate most of the noise that originates from random number generation and the posterior distributions of the calibrated education and alcohol models.

First, uncertainty through stochasticity was explored by running 20 iterations of the microsimulation with unique random number seeds but identical parameter setting. We then calculated the mean alcohol consumption level by group (i.e., education and sex) across one, two, three etc. up to 20 simulation runs. Figure S5 illustrates the deviation of the calculated mean alcohol consumption level in g per day in the year of the policy (2019) by the number of simulation runs taken into account compared to the mean alcohol consumption level across all 20 iterations.

Deviations were overall small with up to -0.06 g per day in males and -0.01 g per day in females. For males, mean consumption levels across education groups stabilised after about 10 iterations with a variation within ± 0.02 g per day. In females, variation was within ± 0.01 g per day after five iterations across all education groups. Considering this overall very small variation across simulation runs, we chose 10 iterations as sufficient to obtain robust model estimates.

When presenting results from the microsimulation in the manuscript (i.e., changes in alcohol use categories and consumption levels by education and sex over time), uncertainty through stochasticity is therefore controlled for by using the average across $n = 10$ iterations with unique random number seeds.

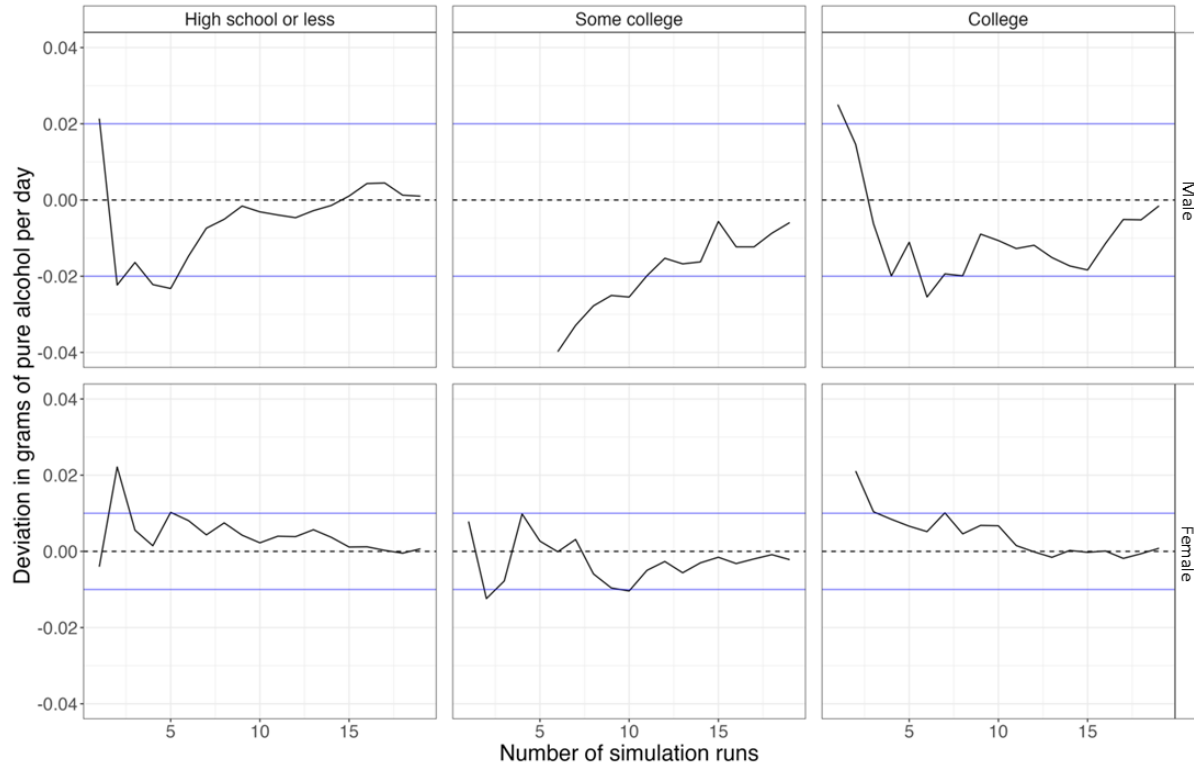


Figure S5. College: college degree or more. Deviation in mean alcohol consumption level in grams of pure alcohol per day in 2019 by number of simulation runs with unique random number seed compared to the mean alcohol consumption level across 20 simulation runs. Model configuration: standard policy setting, scenario 4 (50% price increase in beer and spirits, 10% price increase in wine).

Second, parameter uncertainty was explored by running 150 iterations of the microsimulation with unique combinations of parameters from the posterior distributions of the education and alcohol transition models, as well as beverage-specific own-price consumption elasticities drawn from their population distributions. We calculated the mean alcohol consumption level by group (i.e., education and sex) across one, two, three etc. up to 150 simulation runs. Figure S6 illustrates the deviation of the calculated mean alcohol consumption level in g per day in the year of the policy (2019) by the number of simulation runs taken into account compared to the mean alcohol consumption level across all 150 iterations.

Deviations were within 0.48 g per day in males and 0.19 g per day in females. In both, males and females, mean consumption levels across education groups stabilised after about 60 iterations with a variation within ± 0.02 g per day and ± 0.01 g per day, respectively. Therefore, 60 iterations were chosen to be sufficient for obtaining robust model estimates. We acknowledge that this approach does not account for the total uncertainty associated with the posterior distributions of the calibrated education and alcohol transition models. However, given the overall limited variation observed, we expect that this approach still reflects a reasonable degree of uncertainty

When presenting results from the microsimulation in the manuscript (i.e., changes in alcohol use categories and consumption levels by education and sex over time), uncertainty through parameter uncertainty is displayed in credible intervals (CI) by presenting the minimum and maximum values across $n = 60$ iterations using unique combinations of the calibrated versions of

the education and alcohol models and randomly sampled policy effect estimates. Mean effect estimates were calculated by averaging across these 60 parameter combinations. In combination with $n = 10$ iterations with unique random number seeds to control for stochasticity (see above), we run $n = 600$ model iterations for each policy scenario in total.

Although policy effects on both drinking participation and consumption levels are modelled, we accounted for uncertainty in beverage-specific own-price consumption elasticities only. Thus, we did not model uncertainty in the policy effect of drinking participation, given limited evidence on its presence^{18,20}. Instead, we conducted sensitivity analyses with and without this parameter (see manuscript and sensitivity analyses 4B, 5B, and 6B for results with no participation elasticities, i.e., more conservative setting).

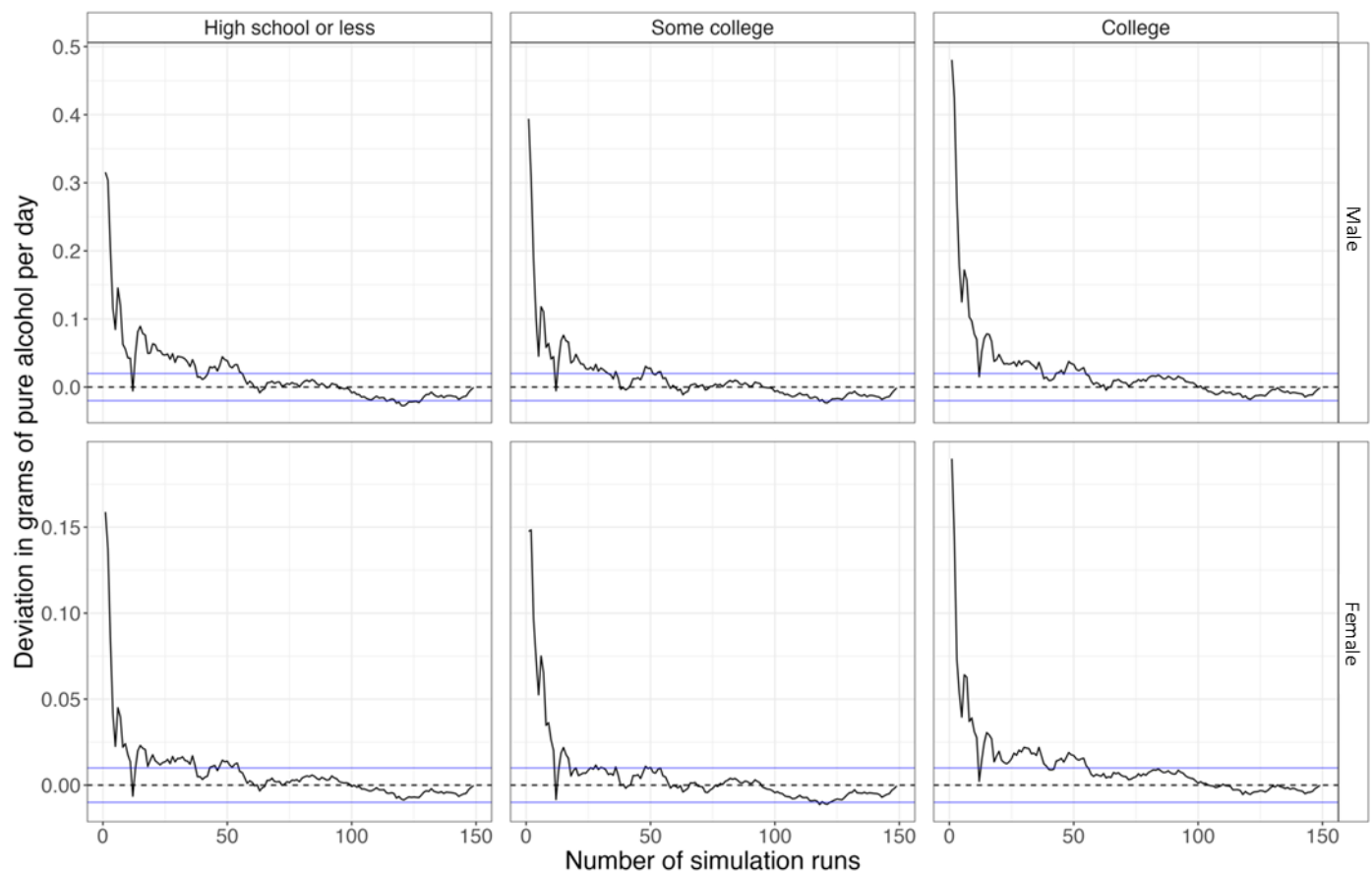


Figure S6. College: college degree or more. Deviation in mean alcohol consumption level in grams of pure alcohol per day in 2019 by number of simulation runs with unique combinations of the calibrated education and alcohol transition models and beverage-specific own-price consumption elasticities drawn from population distribution compared to the mean alcohol consumption level across 150 simulation runs. Model configuration: standard policy setting, scenario 4 (50% price increase in beer and spirits, 10% price increase in wine), random number seed: 1468.

Supplement 4A. Simulated average alcohol consumption levels in the total population (main analysis, standard policy setting).

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
Reference scenario (no policy)					
2000	M	≤High school	14.94	.	.
2000	M	Some college	15.31	.	.
2000	M	≥College	13.26	.	.
2000	F	≤High school	4.40	.	.
2000	F	Some college	5.56	.	.
2000	F	≥College	6.01	.	.
2019	M	≤High school	13.87 (13.83, 13.91)	.	.
2019	M	Some college	13.96 (13.90, 14.00)	.	.
2019	M	≥College	14.91 (14.86, 14.96)	.	.
2019	F	≤High school	5.98 (5.96, 6.01)	.	.
2019	F	Some college	6.63 (6.61, 6.66)	.	.
2019	F	≥College	7.79 (7.77, 7.82)	.	.
Scenario 1: Price increase by 10% for beer, wine, and spirits					
2019	M	≤High school	13.06 (12.92, 13.20)	-0.81 (-0.95, -0.66)	-5.85 (-6.85, -4.77)
2019	M	Some college	13.10 (12.97, 13.25)	-0.86 (-0.99, -0.72)	-6.17 (-7.06, -5.18)
2019	M	≥College	13.96 (13.80, 14.12)	-0.95 (-1.07, -0.83)	-6.40 (-7.15, -5.56)
2019	F	≤High school	5.61 (5.56, 5.67)	-0.37 (-0.42, -0.32)	-6.26 (-7.04, -5.36)
2019	F	Some college	6.20 (6.15, 6.27)	-0.43 (-0.48, -0.37)	-6.44 (-7.22, -5.58)
2019	F	≥College	7.30 (7.23, 7.39)	-0.50 (-0.56, -0.42)	-6.39 (-7.18, -5.37)
Scenario 2: Price increase by 30% for beer, wine, and spirits					
2019	M	≤High school	11.49 (11.08, 11.91)	-2.38 (-2.78, -1.95)	-17.15 (-20.05, -14.05)
2019	M	Some college	11.44 (11.07, 11.85)	-2.52 (-2.88, -2.12)	-18.04 (-20.67, -15.19)
2019	M	≥College	12.12 (11.76, 12.53)	-2.79 (-3.11, -2.43)	-18.71 (-20.86, -16.24)
2019	F	≤High school	4.88 (4.75, 5.04)	-1.10 (-1.24, -0.94)	-18.36 (-20.61, -15.75)
2019	F	Some college	5.38 (5.23, 5.55)	-1.25 (-1.40, -1.09)	-18.88 (-21.09, -16.43)
2019	F	≥College	6.34 (6.15, 6.58)	-1.46 (-1.63, -1.23)	-18.69 (-20.98, -15.75)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	11.71 (11.30, 12.13)	-2.16 (-2.56, -1.72)	-15.55 (-18.47, -12.45)
2019	M	Some college	11.75 (11.38, 12.16)	-2.21 (-2.58, -1.82)	-15.83 (-18.49, -12.99)
2019	M	≥College	12.62 (12.29, 13.02)	-2.29 (-2.61, -1.93)	-15.34 (-17.50, -12.93)
2019	F	≤High school	5.08 (4.94, 5.22)	-0.91 (-1.04, -0.75)	-15.14 (-17.41, -12.59)
2019	F	Some college	5.65 (5.52, 5.81)	-0.97 (-1.11, -0.83)	-14.70 (-16.71, -12.53)
2019	F	≥College	6.76 (6.63, 6.90)	-1.03 (-1.16, -0.90)	-13.24 (-14.87, -11.54)

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	10.41 (9.75, 11.10)	-3.46 (-4.12, -2.76)	-24.92 (-29.69, -19.89)
2019	M	Some college	10.45 (9.84, 11.11)	-3.51 (-4.11, -2.87)	-25.16 (-29.46, -20.54)
2019	M	≥College	11.34 (10.80, 11.95)	-3.57 (-4.09, -3.00)	-23.96 (-27.48, -20.08)
2019	F	≤High school	4.56 (4.34, 4.80)	-1.42 (-1.64, -1.17)	-23.73 (-27.43, -19.61)
2019	F	Some college	5.12 (4.91, 5.36)	-1.50 (-1.72, -1.27)	-22.70 (-25.90, -19.19)
2019	F	≥College	6.24 (6.05, 6.45)	-1.55 (-1.75, -1.34)	-19.89 (-22.42, -17.19)

g per day: grams of pure alcohol per day, CI: Credible interval, minimum and maximum of the simulated policy effect across 60 model iterations, M: Males, F: Females, ≤High school: High school degree or less, ≥College: College degree or more. Standard policy setting: $r = 0.6$, application of beverage-nonspecific own-price participation elasticity.

Supplement 4B. Simulated average alcohol consumption levels in the total population (sensitivity analysis 1, more conservative policy setting).

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
Reference scenario (no policy)					
2000	M	≤High school	14·94	·	·
2000	M	Some college	15·31	·	·
2000	M	≥College	13·26	·	·
2000	F	≤High school	4·40	·	·
2000	F	Some college	5·56	·	·
2000	F	≥College	6·01	·	·
2019	M	≤High school	13·87 (13·83, 13·91)	·	·
2019	M	Some college	13·96 (13·90, 14·00)	·	·
2019	M	≥College	14·91 (14·86, 14·96)	·	·
2019	F	≤High school	5·98 (5·96, 6·01)	·	·
2019	F	Some college	6·63 (6·61, 6·66)	·	·
2019	F	≥College	7·79 (7·77, 7·82)	·	·
Scenario 1: Price increase by 10% for beer, wine, and spirits					
2019	M	≤High school	13·31 (13·17, 13·43)	-0·56 (-0·69, -0·43)	-4·07 (-4·99, -3·09)
2019	M	Some college	13·35 (13·23, 13·50)	-0·61 (-0·72, -0·47)	-4·34 (-5·19, -3·40)
2019	M	≥College	14·24 (14·09, 14·41)	-0·67 (-0·77, -0·55)	-4·48 (-5·20, -3·66)
2019	F	≤High school	5·70 (5·66, 5·76)	-0·28 (-0·33, -0·23)	-4·73 (-5·48, -3·86)
2019	F	Some college	6·31 (6·26, 6·37)	-0·32 (-0·37, -0·26)	-4·82 (-5·56, -3·99)
2019	F	≥College	7·43 (7·36, 7·52)	-0·36 (-0·42, -0·29)	-4·68 (-5·44, -3·72)
Scenario 2: Price increase by 30% for beer, wine, and spirits					
2019	M	≤High school	12·18 (11·79, 12·57)	-1·69 (-2·08, -1·28)	-12·22 (-14·97, -9·26)
2019	M	Some college	12·14 (11·78, 12·55)	-1·82 (-2·17, -1·42)	-13·02 (-15·58, -10·19)
2019	M	≥College	12·91 (12·55, 13·31)	-2·00 (-2·32, -1·64)	-13·45 (-15·60, -10·97)
2019	F	≤High school	5·13 (5·00, 5·29)	-0·85 (-0·98, -0·69)	-14·18 (-16·43, -11·57)
2019	F	Some college	5·67 (5·52, 5·84)	-0·96 (-1·11, -0·79)	-14·47 (-16·69, -11·98)
2019	F	≥College	6·70 (6·52, 6·94)	-1·09 (-1·27, -0·87)	-14·04 (-16·32, -11·15)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	12·31 (11·92, 12·70)	-1·56 (-1·95, -1·16)	-11·28 (-14·04, -8·35)
2019	M	Some college	12·35 (11·99, 12·76)	-1·61 (-1·96, -1·22)	-11·51 (-14·07, -8·73)
2019	M	≥College	13·29 (12·96, 13·69)	-1·62 (-1·94, -1·27)	-10·86 (-13·02, -8·47)
2019	F	≤High school	5·29 (5·16, 5·43)	-0·69 (-0·83, -0·54)	-11·57 (-13·84, -9·05)
2019	F	Some college	5·90 (5·77, 6·05)	-0·73 (-0·86, -0·58)	-11·01 (-13·01, -8·81)
2019	F	≥College	7·06 (6·93, 7·19)	-0·74 (-0·86, -0·60)	-9·46 (-11·09, -7·75)

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	11.31 (10.66, 11.97)	-2.56 (-3.20, -1.89)	-18.49 (-23.08, -13.62)
2019	M	Some college	11.35 (10.75, 12.01)	-2.61 (-3.20, -1.97)	-18.67 (-22.94, -14.07)
2019	M	≥College	12.34 (11.80, 12.97)	-2.57 (-3.10, -1.99)	-17.24 (-20.83, -13.29)
2019	F	≤High school	4.88 (4.66, 5.12)	-1.10 (-1.33, -0.85)	-18.42 (-22.20, -14.25)
2019	F	Some college	5.49 (5.28, 5.73)	-1.14 (-1.36, -0.90)	-17.19 (-20.46, -13.57)
2019	F	≥College	6.68 (6.48, 6.90)	-1.11 (-1.31, -0.89)	-14.23 (-16.86, -11.45)

g per day: grams of pure alcohol per day, CI: Credible interval, minimum and maximum of the simulated policy effect across 60 model iterations, M: Males, F: Females, ≤High school: High school degree or less, ≥College: College degree or more. More conservative policy setting: $r = 0.8$, no application of beverage-nonspecific own-price participation elasticity.

Supplement 4C. Simulated average alcohol consumption levels in the total population (sensitivity analysis 2, less conservative policy setting).

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
Reference scenario (no policy)					
2000	M	≤High school	14.94	.	.
2000	M	Some college	15.31	.	.
2000	M	≥College	13.26	.	.
2000	F	≤High school	4.40	.	.
2000	F	Some college	5.56	.	.
2000	F	≥College	6.01	.	.
2019	M	≤High school	13.87 (13.83, 13.91)	.	.
2019	M	Some college	13.96 (13.90, 14.00)	.	.
2019	M	≥College	14.91 (14.86, 14.96)	.	.
2019	F	≤High school	5.98 (5.96, 6.01)	.	.
2019	F	Some college	6.63 (6.61, 6.66)	.	.
2019	F	≥College	7.79 (7.77, 7.82)	.	.
Scenario 1: Price increase by 10% for beer, wine, and spirits					
2019	M	≤High school	13.01 (12.86, 13.16)	-0.86 (-1.01, -0.69)	-6.17 (-7.27, -5.00)
2019	M	Some college	13.06 (12.92, 13.22)	-0.90 (-1.03, -0.75)	-6.44 (-7.40, -5.39)
2019	M	≥College	13.92 (13.76, 14.09)	-0.99 (-1.11, -0.86)	-6.65 (-7.43, -5.77)
2019	F	≤High school	5.59 (5.55, 5.66)	-0.39 (-0.43, -0.33)	-6.45 (-7.26, -5.51)
2019	F	Some college	6.19 (6.14, 6.26)	-0.44 (-0.49, -0.38)	-6.62 (-7.42, -5.73)
2019	F	≥College	7.28 (7.21, 7.38)	-0.51 (-0.58, -0.43)	-6.60 (-7.44, -5.51)
Scenario 2: Price increase by 30% for beer, wine, and spirits					
2019	M	≤High school	11.36 (10.92, 11.82)	-2.51 (-2.95, -2.04)	-18.10 (-21.26, -14.73)
2019	M	Some college	11.33 (10.94, 11.77)	-2.63 (-3.02, -2.21)	-18.84 (-21.64, -15.80)
2019	M	≥College	12.01 (11.63, 12.44)	-2.90 (-3.23, -2.52)	-19.44 (-21.72, -16.85)
2019	F	≤High school	4.85 (4.71, 5.01)	-1.13 (-1.27, -0.97)	-18.92 (-21.27, -16.21)
2019	F	Some college	5.34 (5.19, 5.52)	-1.29 (-1.44, -1.12)	-19.39 (-21.68, -16.85)
2019	F	≥College	6.29 (6.10, 6.54)	-1.50 (-1.69, -1.26)	-19.30 (-21.72, -16.15)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	11.58 (11.14, 12.04)	-2.29 (-2.73, -1.82)	-16.50 (-19.69, -13.12)
2019	M	Some college	11.64 (11.24, 12.08)	-2.32 (-2.71, -1.90)	-16.61 (-19.45, -13.58)
2019	M	≥College	12.53 (12.18, 12.94)	-2.38 (-2.72, -2.01)	-15.96 (-18.22, -13.44)
2019	F	≤High school	5.05 (4.91, 5.20)	-0.93 (-1.08, -0.77)	-15.62 (-17.98, -12.97)
2019	F	Some college	5.63 (5.50, 5.79)	-1.00 (-1.14, -0.85)	-15.06 (-17.13, -12.83)
2019	F	≥College	6.74 (6.61, 6.88)	-1.05 (-1.19, -0.92)	-13.53 (-15.21, -11.79)

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	10.20 (9.48, 10.95)	-3.67 (-4.39, -2.91)	-26.47 (-31.66, -20.99)
2019	M	Some college	10.27 (9.63, 10.97)	-3.69 (-4.33, -3.00)	-26.42 (-31.01, -21.49)
2019	M	≥College	11.19 (10.64, 11.83)	-3.72 (-4.26, -3.12)	-24.93 (-28.61, -20.87)
2019	F	≤High school	4.52 (4.29, 4.76)	-1.47 (-1.70, -1.21)	-24.50 (-28.34, -20.21)
2019	F	Some college	5.09 (4.87, 5.34)	-1.54 (-1.76, -1.30)	-23.23 (-26.55, -19.64)
2019	F	≥College	6.22 (6.01, 6.43)	-1.58 (-1.78, -1.36)	-20.24 (-22.85, -17.49)

g per day: grams of pure alcohol per day, CI: Credible interval, minimum and maximum of the simulated policy effect across 60 model iterations, M: Males, F: Females, ≤High school: High school degree or less, ≥College: College degree or more. Less conservative policy setting: $r=0.4$, application of beverage-nonspecific own-price participation elasticity.

Supplement 5A. Simulated prevalence of alcohol use categories (main analysis, standard policy setting).

			Abstinence		Alcohol use category 1		Alcohol use category 2		Alcohol use category 3	
Year	Sex	Education	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)
Reference scenario (no policy)										
2000	M	≤High school	28·06	·	61·52	·	4·45	·	5·98	·
2000	M	Some college	21·25	·	67·78	·	4·97	·	6·00	·
2000	M	≥College	18·14	·	73·63	·	5·08	·	3·15	·
2000	F	≤High school	38·60	·	55·96	·	3·37	·	2·08	·
2000	F	Some college	28·22	·	64·64	·	4·59	·	2·55	·
2000	F	≥College	22·17	·	70·04	·	5·50	·	2·29	·
2019	M	≤High school	29·54 (29·48, 29·59)	·	60·40 (60·34, 60·46)	·	4·89 (4·86, 4·93)	·	5·17 (5·15, 5·21)	·
2019	M	Some college	22·50 (22·42, 22·60)	·	67·10 (66·97, 67·19)	·	5·93 (5·89, 5·98)	·	4·46 (4·43, 4·51)	·
2019	M	≥College	19·12 (19·07, 19·19)	·	69·72 (69·64, 69·79)	·	6·54 (6·51, 6·58)	·	4·61 (4·58, 4·64)	·
2019	F	≤High school	31·91 (31·85, 32·01)	·	60·56 (60·48, 60·62)	·	4·28 (4·23, 4·32)	·	3·25 (3·23, 3·28)	·
2019	F	Some college	23·17 (23·10, 23·27)	·	68·18 (68·09, 68·27)	·	5·38 (5·34, 5·44)	·	3·26 (3·24, 3·29)	·
2019	F	≥College	18·92 (18·85, 19·00)	·	70·54 (70·45, 70·62)	·	6·45 (6·41, 6·48)	·	4·09 (4·05, 4·13)	·
Scenario 1: Price increase by 10% for beer, wine, and spirits										
2019	M	≤High school	31·60 (31·54, 31·66)	2·06 (2·05, 2·08)	59·20 (59·04, 59·39)	-1·19 (-1·39, -1·02)	4·34 (4·21, 4·47)	-0·55 (-0·68, -0·43)	4·86 (4·80, 4·92)	-0·32 (-0·37, -0·25)
2019	M	Some college	24·79 (24·71, 24·88)	2·29 (2·27, 2·31)	65·90 (65·65, 66·05)	-1·21 (-1·40, -1·03)	5·22 (5·10, 5·37)	-0·72 (-0·84, -0·56)	4·09 (4·03, 4·17)	-0·37 (-0·43, -0·31)
2019	M	≥College	21·50 (21·43, 21·56)	2·37 (2·35, 2·40)	68·66 (68·40, 68·86)	-1·06 (-1·26, -0·88)	5·75 (5·64, 5·91)	-0·80 (-0·91, -0·66)	4·10 (4·02, 4·20)	-0·51 (-0·59, -0·44)
2019	F	≤High school	33·97 (33·90, 34·05)	2·06 (2·05, 2·08)	59·20 (59·09, 59·31)	-1·37 (-1·46, -1·27)	3·91 (3·83, 3·98)	-0·37 (-0·42, -0·32)	2·92 (2·87, 2·97)	-0·32 (-0·36, -0·28)
2019	F	Some college	25·49 (25·40, 25·58)	2·32 (2·30, 2·34)	66·77 (66·62, 66·89)	-1·41 (-1·54, -1·29)	4·87 (4·79, 4·98)	-0·51 (-0·58, -0·44)	2·87 (2·82, 2·93)	-0·39 (-0·44, -0·35)
2019	F	≥College	21·32 (21·24, 21·41)	2·40 (2·38, 2·43)	69·29 (69·07, 69·45)	-1·25 (-1·44, -1·12)	5·91 (5·83, 6·03)	-0·54 (-0·62, -0·43)	3·48 (3·38, 3·59)	-0·61 (-0·68, -0·53)
Scenario 2: Price increase by 30% for beer, wine, and spirits										
2019	M	≤High school	35·72 (35·65, 35·77)	6·18 (6·16, 6·21)	56·50 (56·04, 56·92)	-3·90 (-4·39, -3·50)	3·75 (3·50, 3·99)	-1·14 (-1·39, -0·90)	4·03 (3·86, 4·27)	-1·14 (-1·32, -0·90)
2019	M	Some college	29·36 (29·28, 29·44)	6·86 (6·81, 6·90)	63·14 (62·59, 63·54)	-3·96 (-4·46, -3·50)	4·15 (3·87, 4·49)	-1·78 (-2·08, -1·42)	3·35 (3·20, 3·54)	-1·11 (-1·26, -0·94)
2019	M	≥College	26·23 (26·14, 26·29)	7·11 (7·07, 7·13)	66·10 (65·58, 66·55)	-3·62 (-4·11, -3·14)	4·44 (4·16, 4·77)	-2·11 (-2·39, -1·78)	3·23 (3·05, 3·45)	-1·38 (-1·57, -1·19)
2019	F	≤High school	38·11 (38·04, 38·17)	6·20 (6·16, 6·23)	55·96 (55·75, 56·15)	-4·61 (-4·80, -4·42)	3·60 (3·49, 3·73)	-0·68 (-0·76, -0·57)	2·33 (2·24, 2·43)	-0·91 (-1·01, -0·81)
2019	F	Some college	30·14 (30·07, 30·23)	6·97 (6·94, 7·02)	63·29 (63·04, 63·52)	-4·89 (-5·17, -4·65)	4·31 (4·18, 4·48)	-1·07 (-1·20, -0·89)	2·26 (2·15, 2·39)	-1·01 (-1·12, -0·89)

Year	Sex	Education	Abstinence		Alcohol use category 1		Alcohol use category 2		Alcohol use category 3	
			Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)
2019	F	≥College	26·11 (26·02, 26·21)	7·19 (7·16, 7·22)	65·89 (65·51, 66·18)	-4·65 (-5·05, -4·38)	5·40 (5·25, 5·62)	-1·06 (-1·21, -0·84)	2·60 (2·43, 2·81)	-1·49 (-1·66, -1·30)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine										
2019	M	≤High school	34·60 (34·55, 34·66)	5·07 (5·05, 5·09)	57·48 (57·00, 57·90)	-2·92 (-3·43, -2·51)	3·84 (3·59, 4·05)	-1·05 (-1·30, -0·84)	4·08 (3·90, 4·40)	-1·10 (-1·28, -0·77)
2019	M	Some college	28·12 (28·05, 28·20)	5·62 (5·58, 5·66)	64·09 (63·50, 64·48)	-3·01 (-3·55, -2·53)	4·37 (4·08, 4·71)	-1·56 (-1·87, -1·20)	3·41 (3·26, 3·66)	-1·05 (-1·21, -0·82)
2019	M	≥College	24·95 (24·86, 25·01)	5·83 (5·79, 5·86)	66·84 (66·25, 67·28)	-2·88 (-3·44, -2·41)	4·84 (4·57, 5·13)	-1·71 (-1·98, -1·40)	3·37 (3·18, 3·71)	-1·24 (-1·43, -0·93)
2019	F	≤High school	36·99 (36·92, 37·05)	5·08 (5·04, 5·10)	56·81 (56·60, 57·03)	-3·75 (-3·96, -3·55)	3·78 (3·68, 3·88)	-0·50 (-0·59, -0·42)	2·42 (2·31, 2·59)	-0·82 (-0·92, -0·64)
2019	F	Some college	28·88 (28·81, 28·98)	5·71 (5·68, 5·75)	64·15 (63·87, 64·38)	-4·03 (-4·34, -3·78)	4·54 (4·42, 4·65)	-0·84 (-0·93, -0·73)	2·42 (2·31, 2·62)	-0·84 (-0·98, -0·65)
2019	F	≥College	24·81 (24·72, 24·91)	5·89 (5·87, 5·92)	66·47 (66·15, 66·73)	-4·08 (-4·33, -3·84)	5·70 (5·63, 5·81)	-0·75 (-0·83, -0·65)	3·02 (2·76, 3·22)	-1·07 (-1·29, -0·86)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine										
2019	M	≤High school	37·60 (37·54, 37·64)	8·06 (8·04, 8·09)	55·53 (54·83, 56·17)	-4·87 (-5·59, -4·24)	3·27 (2·89, 3·71)	-1·62 (-2·00, -1·19)	3·60 (3·33, 3·89)	-1·57 (-1·85, -1·28)
2019	M	Some college	31·44 (31·36, 31·52)	8·94 (8·89, 8·99)	62·03 (61·25, 62·67)	-5·07 (-5·79, -4·34)	3·56 (3·09, 4·07)	-2·37 (-2·86, -1·85)	2·97 (2·74, 3·22)	-1·50 (-1·73, -1·25)
2019	M	≥College	28·40 (28·31, 28·47)	9·28 (9·22, 9·32)	64·71 (63·95, 65·33)	-5·01 (-5·76, -4·36)	4·00 (3·59, 4·48)	-2·55 (-2·97, -2·05)	2·89 (2·63, 3·18)	-1·72 (-1·99, -1·46)
2019	F	≤High school	40·01 (39·94, 40·07)	8·09 (8·05, 8·13)	54·33 (54·03, 54·66)	-6·23 (-6·55, -5·91)	3·53 (3·33, 3·70)	-0·75 (-0·94, -0·57)	2·14 (2·00, 2·27)	-1·11 (-1·24, -0·96)
2019	F	Some college	32·27 (32·20, 32·36)	9·10 (9·07, 9·15)	61·39 (61·04, 61·69)	-6·79 (-7·21, -6·47)	4·24 (4·03, 4·46)	-1·15 (-1·32, -0·92)	2·10 (1·97, 2·29)	-1·16 (-1·32, -0·99)
2019	F	≥College	28·32 (28·22, 28·42)	9·41 (9·37, 9·44)	63·55 (63·20, 63·86)	-6·99 (-7·36, -6·72)	5·58 (5·45, 5·71)	-0·88 (-0·99, -0·75)	2·55 (2·33, 2·83)	-1·54 (-1·72, -1·25)

Alcohol use category 1: up to 20 g per day for females and up to 40 g per day for males, Alcohol use category 2: more than 20 g per day up to 40 g per day for females and more than 40 g per day up to 60 g per day for males, Alcohol use category 3: more than 40 g per day for females and more than 60 g per day for males, CI: Credible interval, minimum and maximum of the simulated policy effect across 60 model iterations, M: Males, F: Females, ≤High school: High school degree or less, ≥College: College degree or more, PP: percentage points. Standard policy setting: $r = 0·6$, application of beverage-nonspecific own-price participation elasticity.

Supplement 5B. Simulated prevalence of alcohol use categories (sensitivity analysis 1, more conservative policy setting).

			Abstinence		Alcohol use category 1		Alcohol use category 2		Alcohol use category 3	
Year	Sex	Education	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)
Reference scenario (no policy)										
2000	M	≤High school	28·06	·	61·52	·	4·45	·	5·98	·
2000	M	Some college	21·25	·	67·78	·	4·97	·	6·00	·
2000	M	≥College	18·14	·	73·63	·	5·08	·	3·15	·
2000	F	≤High school	38·60	·	55·96	·	3·37	·	2·08	·
2000	F	Some college	28·22	·	64·64	·	4·59	·	2·55	·
2000	F	≥College	22·17	·	70·04	·	5·50	·	2·29	·
2019	M	≤High school	29·54 (29·48, 29·59)	·	60·40 (60·34, 60·46)	·	4·89 (4·86, 4·93)	·	5·17 (5·15, 5·21)	·
2019	M	Some college	22·50 (22·42, 22·60)	·	67·10 (66·97, 67·19)	·	5·93 (5·89, 5·98)	·	4·46 (4·43, 4·51)	·
2019	M	≥College	19·12 (19·07, 19·19)	·	69·72 (69·64, 69·79)	·	6·54 (6·51, 6·58)	·	4·61 (4·58, 4·64)	·
2019	F	≤High school	31·91 (31·85, 32·01)	·	60·56 (60·48, 60·62)	·	4·28 (4·23, 4·32)	·	3·25 (3·23, 3·28)	·
2019	F	Some college	23·17 (23·10, 23·27)	·	68·18 (68·09, 68·27)	·	5·38 (5·34, 5·44)	·	3·26 (3·24, 3·29)	·
2019	F	≥College	18·92 (18·85, 19·00)	·	70·54 (70·45, 70·62)	·	6·45 (6·41, 6·48)	·	4·09 (4·05, 4·13)	·
Scenario 1: Price increase by 10% for beer, wine, and spirits										
2019	M	≤High school	29·54 (29·48, 29·59)	0·00 (0·00, 0·00)	61·21 (61·06, 61·39)	0·81 (0·64, 0·98)	4·37 (4·25, 4·48)	-0·53 (-0·64, -0·41)	4·89 (4·83, 4·95)	-0·29 (-0·34, -0·23)
2019	M	Some college	22·50 (22·42, 22·60)	0·00 (0·00, 0·00)	68·13 (67·87, 68·28)	1·03 (0·83, 1·19)	5·24 (5·14, 5·39)	-0·69 (-0·80, -0·54)	4·12 (4·06, 4·19)	-0·34 (-0·39, -0·29)
2019	M	≥College	19·12 (19·07, 19·19)	0·00 (0·00, 0·00)	70·96 (70·72, 71·14)	1·24 (1·05, 1·41)	5·78 (5·68, 5·95)	-0·77 (-0·88, -0·63)	4·13 (4·06, 4·23)	-0·48 (-0·55, -0·41)
2019	F	≤High school	31·91 (31·85, 32·01)	0·00 (0·00, 0·00)	61·25 (61·15, 61·36)	0·68 (0·60, 0·77)	3·91 (3·83, 3·98)	-0·37 (-0·43, -0·32)	2·93 (2·89, 2·99)	-0·31 (-0·34, -0·27)
2019	F	Some college	23·17 (23·10, 23·27)	0·00 (0·00, 0·00)	69·07 (68·92, 69·19)	0·89 (0·79, 1·00)	4·87 (4·79, 4·97)	-0·51 (-0·58, -0·44)	2·89 (2·84, 2·94)	-0·38 (-0·42, -0·33)
2019	F	≥College	18·92 (18·85, 19·00)	0·00 (0·00, 0·00)	71·67 (71·44, 71·82)	1·12 (0·94, 1·24)	5·91 (5·83, 6·03)	-0·55 (-0·62, -0·43)	3·51 (3·42, 3·61)	-0·58 (-0·64, -0·50)
Scenario 2: Price increase by 30% for beer, wine, and spirits										
2019	M	≤High school	29·54 (29·48, 29·59)	0·00 (0·00, 0·00)	62·53 (62·10, 62·93)	2·13 (1·67, 2·51)	3·81 (3·58, 3·96)	-1·09 (-1·31, -0·93)	4·13 (3·97, 4·45)	-1·05 (-1·21, -0·72)
2019	M	Some college	22·50 (22·42, 22·60)	0·00 (0·00, 0·00)	69·83 (69·27, 70·18)	2·73 (2·22, 3·15)	4·23 (3·97, 4·50)	-1·70 (-1·98, -1·42)	3·44 (3·31, 3·68)	-1·03 (-1·16, -0·80)
2019	M	≥College	19·12 (19·07, 19·19)	0·00 (0·00, 0·00)	73·01 (72·54, 73·43)	3·29 (2·82, 3·74)	4·53 (4·26, 4·82)	-2·01 (-2·30, -1·75)	3·33 (3·17, 3·59)	-1·28 (-1·44, -1·04)
2019	F	≤High school	31·91 (31·85, 32·01)	0·00 (0·00, 0·00)	62·12 (61·92, 62·32)	1·56 (1·37, 1·71)	3·60 (3·49, 3·73)	-0·68 (-0·77, -0·57)	2·37 (2·28, 2·46)	-0·87 (-0·96, -0·78)
2019	F	Some college	23·17 (23·10, 23·27)	0·00 (0·00, 0·00)	70·21 (69·95, 70·44)	2·03 (1·80, 2·24)	4·31 (4·18, 4·47)	-1·08 (-1·20, -0·90)	2·31 (2·20, 2·43)	-0·95 (-1·07, -0·84)

Year	Sex	Education	Abstinence		Alcohol use category 1		Alcohol use category 2		Alcohol use category 3	
			Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)
2019	F	≥College	18·92 (18·85, 19·00)	0·00 (0·00, 0·00)	73·01 (72·64, 73·28)	2·47 (2·10, 2·73)	5·38 (5·24, 5·56)	-1·07 (-1·21, -0·90)	2·70 (2·53, 2·92)	-1·39 (-1·56, -1·19)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine										
2019	M	≤High school	29·54 (29·48, 29·59)	0·00 (0·00, 0·00)	62·40 (61·93, 62·81)	2·00 (1·51, 2·40)	3·87 (3·67, 4·00)	-1·02 (-1·22, -0·90)	4·19 (4·00, 4·56)	-0·98 (-1·18, -0·61)
2019	M	Some college	22·50 (22·42, 22·60)	0·00 (0·00, 0·00)	69·56 (68·93, 69·90)	2·45 (1·89, 2·88)	4·43 (4·18, 4·71)	-1·50 (-1·77, -1·21)	3·51 (3·36, 3·81)	-0·95 (-1·11, -0·68)
2019	M	≥College	19·12 (19·07, 19·19)	0·00 (0·00, 0·00)	72·51 (71·93, 72·93)	2·79 (2·21, 3·24)	4·87 (4·64, 5·14)	-1·68 (-1·92, -1·39)	3·50 (3·29, 3·82)	-1·11 (-1·32, -0·82)
2019	F	≤High school	31·91 (31·85, 32·01)	0·00 (0·00, 0·00)	61·87 (61·67, 62·06)	1·30 (1·09, 1·47)	3·76 (3·67, 3·84)	-0·52 (-0·60, -0·46)	2·46 (2·35, 2·66)	-0·78 (-0·88, -0·56)
2019	F	Some college	23·17 (23·10, 23·27)	0·00 (0·00, 0·00)	69·84 (69·54, 70·03)	1·66 (1·38, 1·87)	4·51 (4·40, 4·63)	-0·87 (-0·97, -0·76)	2·48 (2·36, 2·67)	-0·78 (-0·93, -0·61)
2019	F	≥College	18·92 (18·85, 19·00)	0·00 (0·00, 0·00)	72·32 (72·03, 72·62)	1·78 (1·53, 2·04)	5·67 (5·58, 5·77)	-0·79 (-0·88, -0·69)	3·10 (2·86, 3·26)	-0·99 (-1·20, -0·82)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine										
2019	M	≤High school	29·54 (29·48, 29·59)	0·00 (0·00, 0·00)	63·33 (62·72, 63·93)	2·94 (2·30, 3·51)	3·37 (3·03, 3·77)	-1·53 (-1·86, -1·12)	3·76 (3·53, 3·99)	-1·41 (-1·65, -1·17)
2019	M	Some college	22·50 (22·42, 22·60)	0·00 (0·00, 0·00)	70·72 (69·94, 71·25)	3·61 (2·89, 4·23)	3·67 (3·27, 4·16)	-2·26 (-2·68, -1·76)	3·11 (2·91, 3·34)	-1·35 (-1·56, -1·13)
2019	M	≥College	19·12 (19·07, 19·19)	0·00 (0·00, 0·00)	73·74 (73·04, 74·33)	4·02 (3·31, 4·62)	4·10 (3·70, 4·56)	-2·45 (-2·86, -1·97)	3·03 (2·81, 3·30)	-1·58 (-1·80, -1·34)
2019	F	≤High school	31·91 (31·85, 32·01)	0·00 (0·00, 0·00)	62·38 (62·09, 62·68)	1·82 (1·51, 2·11)	3·52 (3·32, 3·68)	-0·76 (-0·94, -0·59)	2·19 (2·06, 2·32)	-1·06 (-1·17, -0·91)
2019	F	Some college	23·17 (23·10, 23·27)	0·00 (0·00, 0·00)	70·45 (70·07, 70·72)	2·27 (1·91, 2·56)	4·22 (4·02, 4·44)	-1·16 (-1·33, -0·94)	2·16 (2·03, 2·33)	-1·11 (-1·26, -0·94)
2019	F	≥College	18·92 (18·85, 19·00)	0·00 (0·00, 0·00)	72·92 (72·57, 73·26)	2·38 (2·04, 2·68)	5·54 (5·42, 5·67)	-0·91 (-1·03, -0·79)	2·62 (2·40, 2·93)	-1·47 (-1·66, -1·15)

Alcohol use category 1: up to 20 g per day for females and up to 40 g per day for males, Alcohol use category 2: more than 20 g per day up to 40 g per day for females and more than 40 g per day up to 60 g per day for males, Alcohol use category 3: more than 40 g per day for females and more than 60 g per day for males, CI: Credible interval, minimum and maximum of the simulated policy effect across 60 model iterations, M: Males, F: Females, ≤High school: High school degree or less, ≥College: College degree or more, PP: percentage points. More conservative policy setting: $r=0·8$, no application of beverage-nonspecific own-price participation elasticity.

Supplement 5C. Simulated prevalence of alcohol use categories (sensitivity analysis 2, less conservative policy setting).

			Abstinence		Alcohol use category 1		Alcohol use category 2		Alcohol use category 3	
Year	Sex	Education	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)
Reference scenario (no policy)										
2000	M	≤High school	28·06	·	61·52	·	4·45	·	5·98	·
2000	M	Some college	21·25	·	67·78	·	4·97	·	6·00	·
2000	M	≥College	18·14	·	73·63	·	5·08	·	3·15	·
2000	F	≤High school	38·60	·	55·96	·	3·37	·	2·08	·
2000	F	Some college	28·22	·	64·64	·	4·59	·	2·55	·
2000	F	≥College	22·17	·	70·04	·	5·50	·	2·29	·
2019	M	≤High school	29·54 (29·48, 29·59)	·	60·40 (60·34, 60·46)	·	4·89 (4·86, 4·93)	·	5·17 (5·15, 5·21)	·
2019	M	Some college	22·50 (22·42, 22·60)	·	67·10 (66·97, 67·19)	·	5·93 (5·89, 5·98)	·	4·46 (4·43, 4·51)	·
2019	M	≥College	19·12 (19·07, 19·19)	·	69·72 (69·64, 69·79)	·	6·54 (6·51, 6·58)	·	4·61 (4·58, 4·64)	·
2019	F	≤High school	31·91 (31·85, 32·01)	·	60·56 (60·48, 60·62)	·	4·28 (4·23, 4·32)	·	3·25 (3·23, 3·28)	·
2019	F	Some college	23·17 (23·10, 23·27)	·	68·18 (68·09, 68·27)	·	5·38 (5·34, 5·44)	·	3·26 (3·24, 3·29)	·
2019	F	≥College	18·92 (18·85, 19·00)	·	70·54 (70·45, 70·62)	·	6·45 (6·41, 6·48)	·	4·09 (4·05, 4·13)	·
Scenario 1: Price increase by 10% for beer, wine, and spirits										
2019	M	≤High school	31·60 (31·54, 31·66)	2·06 (2·05, 2·08)	59·25 (59·07, 59·46)	-1·14 (-1·35, - 0·96)	4·32 (4·18, 4·45)	-0·58 (-0·71, - 0·44)	4·83 (4·77, 4·90)	-0·34 (-0·41, - 0·27)
2019	M	Some college	24·79 (24·71, 24·88)	2·29 (2·27, 2·31)	65·95 (65·69, 66·12)	-1·16 (-1·36, - 0·96)	5·19 (5·07, 5·35)	-0·74 (-0·87, - 0·59)	4·07 (4·00, 4·15)	-0·40 (-0·46, - 0·33)
2019	M	≥College	21·50 (21·43, 21·56)	2·37 (2·35, 2·40)	68·72 (68·46, 68·93)	-1·00 (-1·21, - 0·81)	5·72 (5·60, 5·90)	-0·83 (-0·95, - 0·68)	4·06 (3·98, 4·18)	-0·55 (-0·63, - 0·46)
2019	F	≤High school	33·97 (33·90, 34·05)	2·06 (2·05, 2·08)	59·21 (59·10, 59·32)	-1·36 (-1·45, - 1·26)	3·91 (3·83, 3·99)	-0·37 (-0·42, - 0·31)	2·91 (2·86, 2·96)	-0·34 (-0·38, - 0·29)
2019	F	Some college	25·49 (25·40, 25·58)	2·32 (2·30, 2·34)	66·78 (66·63, 66·91)	-1·40 (-1·53, - 1·27)	4·88 (4·80, 4·98)	-0·51 (-0·57, - 0·43)	2·85 (2·80, 2·91)	-0·41 (-0·46, - 0·36)
2019	F	≥College	21·32 (21·24, 21·41)	2·40 (2·38, 2·43)	69·32 (69·09, 69·49)	-1·22 (-1·43, - 1·08)	5·92 (5·83, 6·04)	-0·54 (-0·62, - 0·42)	3·45 (3·35, 3·56)	-0·64 (-0·72, - 0·55)
Scenario 2: Price increase by 30% for beer, wine, and spirits										
2019	M	≤High school	35·72 (35·65, 35·77)	6·18 (6·16, 6·21)	56·63 (56·14, 57·08)	-3·77 (-4·29, - 3·33)	3·70 (3·43, 4·00)	-1·19 (-1·46, - 0·89)	3·95 (3·76, 4·16)	-1·22 (-1·42, - 1·00)
2019	M	Some college	29·36 (29·28, 29·44)	6·86 (6·81, 6·90)	63·29 (62·71, 63·70)	-3·81 (-4·34, - 3·32)	4·07 (3·77, 4·45)	-1·86 (-2·18, - 1·47)	3·28 (3·12, 3·46)	-1·18 (-1·35, - 1·01)
2019	M	≥College	26·23 (26·14, 26·29)	7·11 (7·07, 7·13)	66·27 (65·73, 66·73)	-3·45 (-3·98, - 2·96)	4·36 (4·08, 4·70)	-2·19 (-2·47, - 1·85)	3·14 (2·95, 3·37)	-1·47 (-1·67, - 1·27)
2019	F	≤High school	38·11 (38·04, 38·17)	6·20 (6·16, 6·23)	55·98 (55·78, 56·18)	-4·58 (-4·77, - 4·39)	3·60 (3·49, 3·74)	-0·68 (-0·76, - 0·56)	2·30 (2·20, 2·41)	-0·94 (-1·04, - 0·83)
2019	F	Some college	30·14 (30·07, 30·23)	6·97 (6·94, 7·02)	63·33 (63·07, 63·56)	-4·86 (-5·14, - 4·61)	4·32 (4·18, 4·49)	-1·07 (-1·20, - 0·88)	2·21 (2·10, 2·35)	-1·05 (-1·17, - 0·92)

Year	Sex	Education	Abstinence		Alcohol use category 1		Alcohol use category 2		Alcohol use category 3	
			Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)	Prevalence in % (CI)	Change in PP (CI)
2019	F	≥College	26·11 (26·02, 26·21)	7·19 (7·16, 7·22)	65·96 (65·55, 66·26)	-4·58 (-5·02, -4·30)	5·41 (5·25, 5·66)	-1·04 (-1·20, -0·80)	2·52 (2·35, 2·74)	-1·57 (-1·75, -1·38)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine										
2019	M	≤High school	34·60 (34·55, 34·66)	5·07 (5·05, 5·09)	57·61 (57·10, 58·07)	-2·79 (-3·32, -2·34)	3·80 (3·52, 4·07)	-1·10 (-1·37, -0·83)	3·99 (3·80, 4·26)	-1·18 (-1·38, -0·91)
2019	M	Some college	28·12 (28·05, 28·20)	5·62 (5·58, 5·66)	64·23 (63·62, 64·66)	-2·87 (-3·43, -2·36)	4·31 (3·99, 4·69)	-1·62 (-1·96, -1·22)	3·34 (3·17, 3·55)	-1·13 (-1·30, -0·93)
2019	M	≥College	24·95 (24·86, 25·01)	5·83 (5·79, 5·86)	66·98 (66·39, 67·44)	-2·75 (-3·30, -2·25)	4·79 (4·51, 5·12)	-1·75 (-2·04, -1·41)	3·28 (3·08, 3·59)	-1·33 (-1·53, -1·05)
2019	F	≤High school	36·99 (36·92, 37·05)	5·08 (5·04, 5·10)	56·83 (56·61, 57·05)	-3·73 (-3·94, -3·52)	3·79 (3·69, 3·90)	-0·49 (-0·57, -0·40)	2·39 (2·28, 2·53)	-0·85 (-0·96, -0·70)
2019	F	Some college	28·88 (28·81, 28·98)	5·71 (5·68, 5·75)	64·18 (63·89, 64·40)	-4·00 (-4·32, -3·76)	4·56 (4·44, 4·69)	-0·82 (-0·92, -0·69)	2·38 (2·27, 2·59)	-0·88 (-1·02, -0·69)
2019	F	≥College	24·81 (24·72, 24·91)	5·89 (5·87, 5·92)	66·50 (66·17, 66·76)	-4·05 (-4·31, -3·82)	5·74 (5·65, 5·84)	-0·71 (-0·80, -0·62)	2·95 (2·70, 3·18)	-1·14 (-1·35, -0·90)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine										
2019	M	≤High school	37·60 (37·54, 37·64)	8·06 (8·04, 8·09)	55·74 (54·97, 56·46)	-4·66 (-5·45, -3·95)	3·20 (2·76, 3·66)	-1·70 (-2·13, -1·24)	3·47 (3·17, 3·80)	-1·71 (-2·01, -1·37)
2019	M	Some college	31·44 (31·36, 31·52)	8·94 (8·89, 8·99)	62·25 (61·41, 62·95)	-4·86 (-5·64, -4·06)	3·47 (2·95, 4·00)	-2·47 (-3·00, -1·91)	2·84 (2·60, 3·13)	-1·62 (-1·87, -1·34)
2019	M	≥College	28·40 (28·31, 28·47)	9·28 (9·22, 9·32)	64·91 (64·11, 65·55)	-4·81 (-5·59, -4·14)	3·93 (3·50, 4·42)	-2·62 (-3·06, -2·11)	2·76 (2·48, 3·09)	-1·85 (-2·14, -1·55)
2019	F	≤High school	40·01 (39·94, 40·07)	8·09 (8·05, 8·13)	54·37 (54·05, 54·70)	-6·20 (-6·53, -5·87)	3·54 (3·34, 3·71)	-0·74 (-0·93, -0·56)	2·09 (1·95, 2·24)	-1·15 (-1·29, -0·99)
2019	F	Some college	32·27 (32·20, 32·36)	9·10 (9·07, 9·15)	61·42 (61·06, 61·72)	-6·76 (-7·19, -6·44)	4·25 (4·05, 4·47)	-1·13 (-1·31, -0·90)	2·06 (1·91, 2·25)	-1·21 (-1·37, -1·02)
2019	F	≥College	28·32 (28·22, 28·42)	9·41 (9·37, 9·44)	63·57 (63·22, 63·88)	-6·97 (-7·34, -6·69)	5·61 (5·48, 5·74)	-0·85 (-0·96, -0·72)	2·50 (2·27, 2·76)	-1·59 (-1·78, -1·32)

Alcohol use category 1: up to 20 g per day for females and up to 40 g per day for males, Alcohol use category 2: more than 20 g per day up to 40 g per day for females and more than 40 g per day up to 60 g per day for males, Alcohol use category 3: more than 40 g per day for females and more than 60 g per day for males, CI: Credible interval, minimum and maximum of the simulated policy effect across 60 model iterations, M: Males, F: Females, ≤High school: High school degree or less, ≥College: College degree or more, PP: percentage points. Less conservative policy setting: $r=0·4$, application of beverage-nonspecific own-price participation elasticity.

Supplement 6A. Simulated average consumption levels by alcohol use category as of 2018 (main analysis, standard policy setting).

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
Alcohol use category 1					
Reference scenario (no policy)					
2018	M	≤High school	10·19 (10·17, 10·21)	·	·
2018	M	Some college	10·00 (9·97, 10·03)	·	·
2018	M	≥College	10·93 (10·91, 10·96)	·	·
2018	F	≤High school	4·10 (4·09, 4·10)	·	·
2018	F	Some college	4·40 (4·38, 4·42)	·	·
2018	F	≥College	5·09 (5·07, 5·10)	·	·
2019	M	≤High school	9·75 (9·73, 9·78)	·	·
2019	M	Some college	9·53 (9·51, 9·56)	·	·
2019	M	≥College	10·37 (10·34, 10·40)	·	·
2019	F	≤High school	3·94 (3·93, 3·95)	·	·
2019	F	Some college	4·22 (4·20, 4·23)	·	·
2019	F	≥College	4·87 (4·85, 4·88)	·	·
Scenario 1: Price increase by 10% for beer, wine, and spirits					
2019	M	≤High school	8·95 (8·82, 9·07)	-0·80 (-0·91, -0·69)	-8·25 (-9·40, -7·03)
2019	M	Some college	8·74 (8·63, 8·86)	-0·80 (-0·89, -0·69)	-8·35 (-9·37, -7·23)
2019	M	≥College	9·50 (9·42, 9·62)	-0·86 (-0·95, -0·76)	-8·34 (-9·19, -7·36)
2019	F	≤High school	3·59 (3·56, 3·63)	-0·34 (-0·38, -0·30)	-8·68 (-9·54, -7·67)
2019	F	Some college	3·85 (3·81, 3·90)	-0·37 (-0·40, -0·33)	-8·68 (-9·46, -7·70)
2019	F	≥College	4·45 (4·41, 4·50)	-0·42 (-0·46, -0·36)	-8·58 (-9·46, -7·36)
Scenario 2: Price increase by 30% for beer, wine, and spirits					
2019	M	High school or less	7·43 (7·10, 7·77)	-2·32 (-2·63, -1·98)	-23·78 (-27·02, -20·30)
2019	M	Some college	7·24 (6·96, 7·55)	-2·29 (-2·56, -1·99)	-24·03 (-26·88, -20·88)
2019	M	≥College	7·88 (7·64, 8·17)	-2·49 (-2·74, -2·20)	-24·00 (-26·38, -21·23)
2019	F	High school or less	2·95 (2·86, 3·06)	-0·98 (-1·08, -0·87)	-24·96 (-27·38, -22·17)
2019	F	Some college	3·16 (3·07, 3·28)	-1·05 (-1·15, -0·94)	-24·98 (-27·17, -22·23)
2019	F	≥College	3·67 (3·55, 3·83)	-1·20 (-1·32, -1·03)	-24·67 (-27·09, -21·24)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	7·70 (7·36, 8·04)	-2·05 (-2·37, -1·71)	-21·05 (-24·34, -17·55)
2019	M	Some college	7·56 (7·28, 7·88)	-1·97 (-2·25, -1·67)	-20·67 (-23·57, -17·51)
2019	M	≥College	8·36 (8·12, 8·64)	-2·01 (-2·26, -1·74)	-19·39 (-21·80, -16·77)
2019	F	≤High school	3·14 (3·04, 3·24)	-0·80 (-0·90, -0·69)	-20·29 (-22·76, -17·56)

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
2019	F	Some college	3·40 (3·30, 3·50)	-0·82 (-0·91, -0·72)	-19·44 (-21·61, -17·12)
2019	F	≥College	4·00 (3·92, 4·09)	-0·87 (-0·95, -0·78)	-17·85 (-19·40, -16·01)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	6·53 (6·01, 7·07)	-3·22 (-3·72, -2·68)	-33·00 (-38·22, -27·48)
2019	M	Some college	6·46 (6·02, 6·95)	-3·07 (-3·51, -2·60)	-32·20 (-36·81, -27·24)
2019	M	≥College	7·28 (6·89, 7·73)	-3·09 (-3·49, -2·66)	-29·77 (-33·61, -25·61)
2019	F	≤High school	2·71 (2·55, 2·88)	-1·23 (-1·38, -1·06)	-31·18 (-35·10, -26·89)
2019	F	Some college	2·97 (2·83, 3·13)	-1·25 (-1·38, -1·09)	-29·57 (-32·88, -25·93)
2019	F	≥College	3·57 (3·45, 3·71)	-1·30 (-1·42, -1·16)	-26·63 (-29·14, -23·81)
Alcohol use category 2					
Reference scenario (no policy)					
2000	M	≤High school	47·81 (47·77, 47·84)	·	·
2000	M	Some college	47·69 (47·63, 47·74)	·	·
2000	M	≥College	47·21 (47·17, 47·24)	·	·
2000	F	≤High school	27·16 (27·13, 27·19)	·	·
2000	F	Some college	26·89 (26·85, 26·93)	·	·
2000	F	≥College	26·67 (26·63, 26·70)	·	·
2019	M	≤High school	48·76 (48·68, 48·87)	·	·
2019	M	Some college	48·67 (48·58, 48·79)	·	·
2019	M	≥College	48·09 (48·00, 48·19)	·	·
2019	F	≤High school	28·21 (28·13, 28·30)	·	·
2019	F	Some college	28·08 (27·99, 28·17)	·	·
2019	F	≥College	27·50 (27·42, 27·57)	·	·
Scenario 1: Price increase by 10% for beer, wine, and spirits					
2019	M	≤High school	46·60 (46·12, 47·15)	-2·16 (-2·67, -1·63)	-4·44 (-5·47, -3·34)
2019	M	Some college	46·43 (46·07, 46·99)	-2·24 (-2·68, -1·77)	-4·61 (-5·49, -3·62)
2019	M	≥College	45·80 (45·40, 46·22)	-2·29 (-2·66, -1·89)	-4·75 (-5·54, -3·93)
2019	F	≤High school	26·80 (26·52, 27·06)	-1·41 (-1·63, -1·17)	-5·01 (-5·78, -4·13)
2019	F	Some college	26·67 (26·43, 26·93)	-1·41 (-1·62, -1·17)	-5·03 (-5·78, -4·16)
2019	F	≥College	26·16 (25·94, 26·49)	-1·34 (-1·56, -1·06)	-4·88 (-5·66, -3·85)
Scenario 2: Price increase by 30% for beer, wine, and spirits					
2019	M	≤High school	42·26 (40·76, 43·88)	-6·51 (-8·03, -4·90)	-13·34 (-16·46, -10·05)
2019	M	Some college	41·93 (40·69, 43·44)	-6·74 (-8·05, -5·31)	-13·85 (-16·52, -10·90)
2019	M	≥College	41·22 (40·06, 42·42)	-6·87 (-8·00, -5·70)	-14·29 (-16·65, -11·84)
2019	F	≤High school	23·97 (23·27, 24·72)	-4·25 (-4·89, -3·50)	-15·05 (-17·32, -12·41)
2019	F	Some college	23·84 (23·20, 24·56)	-4·24 (-4·86, -3·51)	-15·09 (-17·33, -12·51)

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
2019	F	≥College	23·47 (22·83, 24·37)	-4·03 (-4·68, -3·17)	-14·65 (-17·03, -11·52)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	42·71 (41·21, 44·32)	-6·05 (-7·57, -4·46)	-12·41 (-15·53, -9·14)
2019	M	Some college	42·68 (41·43, 44·17)	-5·99 (-7·31, -4·59)	-12·31 (-15·00, -9·41)
2019	M	≥College	42·44 (41·35, 43·59)	-5·65 (-6·71, -4·52)	-11·75 (-13·95, -9·40)
2019	F	≤High school	24·81 (24·17, 25·54)	-3·40 (-4·04, -2·69)	-12·06 (-14·33, -9·52)
2019	F	Some college	24·87 (24·30, 25·52)	-3·20 (-3·77, -2·58)	-11·41 (-13·42, -9·19)
2019	F	≥College	24·86 (24·42, 25·33)	-2·64 (-3·08, -2·19)	-9·60 (-11·20, -7·95)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	38·80 (36·27, 41·47)	-9·96 (-12·52, -7·31)	-20·43 (-25·66, -14·99)
2019	M	Some college	38·91 (36·79, 41·34)	-9·76 (-11·95, -7·41)	-20·05 (-24·52, -15·21)
2019	M	≥College	39·06 (37·29, 40·94)	-9·03 (-10·77, -7·18)	-18·77 (-22·41, -14·92)
2019	F	≤High school	22·82 (21·75, 24·01)	-5·39 (-6·47, -4·21)	-19·12 (-22·91, -14·92)
2019	F	Some college	23·08 (22·16, 24·13)	-5·00 (-5·92, -3·97)	-17·80 (-21·09, -14·12)
2019	F	≥College	23·57 (22·86, 24·31)	-3·93 (-4·64, -3·20)	-14·31 (-16·86, -11·63)
Alcohol use category 3					
Reference scenario (no policy)					
2000	M	≤High school	104·15 (103·87, 104·41)	·	·
2000	M	Some college	99·42 (99·15, 99·60)	·	·
2000	M	≥College	93·35 (93·03, 93·63)	·	·
2000	F	≤High school	71·89 (71·61, 72·20)	·	·
2000	F	Some college	67·50 (67·11, 67·83)	·	·
2000	F	≥College	61·34 (61·11, 61·54)	·	·
2019	M	≤High school	101·84 (101·65, 102·11)	·	·
2019	M	Some college	96·37 (96·07, 96·64)	·	·
2019	M	≥College	91·00 (90·65, 91·29)	·	·
2019	F	≤High school	70·06 (69·72, 70·39)	·	·
2019	F	Some college	65·19 (64·85, 65·47)	·	·
2019	F	≥College	59·79 (59·55, 60·02)	·	·
Scenario 1: Price increase by 10% for beer, wine, and spirits					
2019	M	≤High school	98·15 (97·45, 99·11)	-3·69 (-4·50, -2·82)	-3·63 (-4·41, -2·76)
2019	M	Some college	92·71 (91·94, 93·61)	-3·66 (-4·35, -2·90)	-3·80 (-4·52, -3·00)
2019	M	≥College	87·41 (86·83, 88·39)	-3·59 (-4·28, -2·89)	-3·94 (-4·69, -3·17)
2019	F	≤High school	67·15 (66·44, 67·91)	-2·92 (-3·44, -2·37)	-4·16 (-4·90, -3·39)
2019	F	Some college	62·46 (61·96, 63·11)	-2·73 (-3·21, -2·25)	-4·19 (-4·93, -3·45)
2019	F	≥College	57·36 (56·79, 57·88)	-2·44 (-2·85, -2·02)	-4·08 (-4·78, -3·36)

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
Scenario 2: Price increase by 30% for beer, wine, and spirits					
2019	M	≤High school	90·63 (88·30, 93·38)	-11·21 (-13·68, -8·54)	-11·00 (-13·41, -8·38)
2019	M	Some college	85·27 (83·10, 87·71)	-11·11 (-13·19, -8·79)	-11·52 (-13·69, -9·11)
2019	M	≥College	80·12 (78·26, 82·54)	-10·88 (-12·98, -8·75)	-11·95 (-14·23, -9·58)
2019	F	≤High school	61·24 (59·77, 62·94)	-8·82 (-10·39, -7·18)	-12·59 (-14·80, -10·25)
2019	F	Some college	56·93 (55·48, 58·43)	-8·26 (-9·72, -6·80)	-12·68 (-14·90, -10·43)
2019	F	≥College	52·42 (51·02, 53·80)	-7·37 (-8·62, -6·09)	-12·33 (-14·46, -10·17)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	91·25 (88·93, 93·96)	-10·59 (-13·06, -7·96)	-10·40 (-12·80, -7·81)
2019	M	Some college	86·13 (83·99, 88·55)	-10·24 (-12·32, -7·96)	-10·63 (-12·79, -8·24)
2019	M	≥College	81·61 (79·85, 83·95)	-9·39 (-11·39, -7·33)	-10·32 (-12·49, -8·03)
2019	F	≤High school	62·51 (61·04, 64·02)	-7·55 (-9·05, -5·96)	-10·78 (-12·89, -8·52)
2019	F	Some college	58·59 (57·24, 60·04)	-6·60 (-7·95, -5·20)	-10·12 (-12·20, -7·98)
2019	F	≥College	54·56 (53·49, 55·69)	-5·23 (-6·28, -4·16)	-8·75 (-10·50, -6·95)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	84·22 (80·21, 88·73)	-17·62 (-21·77, -13·20)	-17·30 (-21·35, -12·95)
2019	M	Some college	79·44 (75·90, 83·40)	-16·94 (-20·41, -13·10)	-17·57 (-21·19, -13·58)
2019	M	≥College	75·70 (72·61, 79·43)	-15·30 (-18·63, -11·85)	-16·81 (-20·42, -12·98)
2019	F	≤High school	57·81 (55·37, 60·40)	-12·25 (-14·72, -9·58)	-17·49 (-20·98, -13·69)
2019	F	Some college	54·68 (52·45, 56·98)	-10·51 (-12·74, -8·19)	-16·13 (-19·55, -12·57)
2019	F	≥College	51·74 (50·00, 53·56)	-8·06 (-9·77, -6·28)	-13·47 (-16·34, -10·49)

g per day: grams of pure alcohol per day, CI: Credible interval, minimum and maximum of the simulated policy effect across 60 model iterations, Alcohol use category 1: up to 20 g per day for females and up to 40 g per day for males, Alcohol use category 2: more than 20 g per day up to 40 g per day for females and more than 40 g per day up to 60 g per day for males, Alcohol use category 3: more than 40 g per day for females and more than 60 g per day for males, M: Males, F: Females, ≤High school: High school degree or less, ≥College: College degree or more. Standard policy setting: $r=0·6$, application of beverage-nonspecific own-price participation elasticity.

Supplement 6B. Simulated average consumption levels by alcohol use category as of 2018 (sensitivity analysis 1, more conservative policy setting).

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
Alcohol use category 1					
Reference scenario (no policy)					
2018	M	≤High school	10·19 (10·17, 10·21)	·	·
2018	M	Some college	10·00 (9·97, 10·03)	·	·
2018	M	≥College	10·93 (10·91, 10·96)	·	·
2018	F	≤High school	4·10 (4·09, 4·10)	·	·
2018	F	Some college	4·40 (4·38, 4·42)	·	·
2018	F	≥College	5·09 (5·07, 5·10)	·	·
2019	M	≤High school	9·75 (9·73, 9·78)	·	·
2019	M	Some college	9·53 (9·51, 9·56)	·	·
2019	M	≥College	10·37 (10·34, 10·40)	·	·
2019	F	≤High school	3·94 (3·93, 3·95)	·	·
2019	F	Some college	4·22 (4·20, 4·23)	·	·
2019	F	≥College	4·87 (4·85, 4·88)	·	·
Scenario 1: Price increase by 10% for beer, wine, and spirits					
2019	M	≤High school	9·26 (9·13, 9·38)	-0·49 (-0·60, -0·37)	-5·05 (-6·20, -3·81)
2019	M	Some college	9·04 (8·93, 9·16)	-0·49 (-0·59, -0·39)	-5·17 (-6·21, -4·03)
2019	M	≥College	9·83 (9·75, 9·95)	-0·53 (-0·62, -0·43)	-5·15 (-6·01, -4·16)
2019	F	≤High school	3·71 (3·67, 3·76)	-0·22 (-0·26, -0·18)	-5·67 (-6·57, -4·63)
2019	F	Some college	3·98 (3·93, 4·03)	-0·24 (-0·27, -0·20)	-5·66 (-6·47, -4·64)
2019	F	≥College	4·60 (4·56, 4·66)	-0·27 (-0·31, -0·21)	-5·49 (-6·36, -4·25)
Scenario 2: Price increase by 30% for beer, wine, and spirits					
2019	M	≤High school	8·27 (7·92, 8·64)	-1·48 (-1·81, -1·11)	-15·14 (-18·61, -11·42)
2019	M	Some college	8·05 (7·75, 8·39)	-1·48 (-1·77, -1·16)	-15·51 (-18·62, -12·10)
2019	M	≥College	8·77 (8·51, 9·09)	-1·60 (-1·87, -1·29)	-15·44 (-18·03, -12·47)
2019	F	≤High school	3·27 (3·16, 3·39)	-0·67 (-0·78, -0·55)	-17·00 (-19·71, -13·89)
2019	F	Some college	3·50 (3·39, 3·64)	-0·72 (-0·82, -0·59)	-16·97 (-19·40, -13·91)
2019	F	≥College	4·06 (3·94, 4·24)	-0·80 (-0·93, -0·62)	-16·48 (-19·08, -12·76)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	8·40 (8·05, 8·77)	-1·35 (-1·68, -0·99)	-13·80 (-17·28, -10·14)
2019	M	Some college	8·25 (7·94, 8·58)	-1·29 (-1·58, -0·97)	-13·49 (-16·61, -10·15)
2019	M	≥College	9·11 (8·85, 9·42)	-1·26 (-1·53, -0·97)	-12·13 (-14·72, -9·31)
2019	F	≤High school	3·40 (3·30, 3·52)	-0·53 (-0·64, -0·41)	-13·49 (-16·20, -10·50)

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
2019	F	Some college	3·69 (3·59, 3·80)	-0·53 (-0·62, -0·42)	-12·54 (-14·81, -9·98)
2019	F	≥College	4·34 (4·26, 4·44)	-0·52 (-0·61, -0·43)	-10·74 (-12·49, -8·74)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	7·55 (6·97, 8·15)	-2·20 (-2·76, -1·61)	-22·56 (-28·35, -16·46)
2019	M	Some college	7·45 (6·95, 7·99)	-2·08 (-2·57, -1·55)	-21·80 (-27·00, -16·26)
2019	M	≥College	8·39 (7·95, 8·89)	-1·98 (-2·43, -1·50)	-19·10 (-23·43, -14·44)
2019	F	≤High school	3·10 (2·92, 3·29)	-0·84 (-1·02, -0·64)	-21·32 (-25·83, -16·36)
2019	F	Some college	3·40 (3·23, 3·58)	-0·82 (-0·98, -0·64)	-19·43 (-23·21, -15·22)
2019	F	≥College	4·09 (3·95, 4·25)	-0·78 (-0·92, -0·62)	-15·99 (-18·91, -12·73)
Alcohol use category 2					
Reference scenario (no policy)					
2000	M	≤High school	47·81 (47·77, 47·84)	·	·
2000	M	Some college	47·69 (47·63, 47·74)	·	·
2000	M	≥College	47·21 (47·17, 47·24)	·	·
2000	F	≤High school	27·16 (27·13, 27·19)	·	·
2000	F	Some college	26·89 (26·85, 26·93)	·	·
2000	F	≥College	26·67 (26·63, 26·70)	·	·
2019	M	≤High school	48·76 (48·68, 48·87)	·	·
2019	M	Some college	48·67 (48·58, 48·79)	·	·
2019	M	≥College	48·09 (48·00, 48·19)	·	·
2019	F	≤High school	28·21 (28·13, 28·30)	·	·
2019	F	Some college	28·08 (27·99, 28·17)	·	·
2019	F	≥College	27·50 (27·42, 27·57)	·	·
Scenario 1: Price increase by 10% for beer, wine, and spirits					
2019	M	≤High school	46·86 (46·44, 47·35)	-1·90 (-2·35, -1·43)	-3·90 (-4·82, -2·92)
2019	M	Some college	46·67 (46·34, 47·18)	-2·01 (-2·40, -1·57)	-4·12 (-4·93, -3·23)
2019	M	≥College	46·04 (45·67, 46·42)	-2·05 (-2·39, -1·69)	-4·26 (-4·98, -3·51)
2019	F	≤High school	26·89 (26·62, 27·14)	-1·32 (-1·53, -1·08)	-4·68 (-5·41, -3·84)
2019	F	Some college	26·76 (26·54, 27·02)	-1·31 (-1·51, -1·09)	-4·68 (-5·40, -3·87)
2019	F	≥College	26·28 (26·08, 26·58)	-1·23 (-1·43, -0·97)	-4·46 (-5·20, -3·52)
Scenario 2: Price increase by 30% for beer, wine, and spirits					
2019	M	≤High school	43·06 (41·73, 44·50)	-5·71 (-7·06, -4·28)	-11·70 (-14·46, -8·76)
2019	M	Some college	42·65 (41·54, 44·04)	-6·02 (-7·21, -4·72)	-12·36 (-14·79, -9·68)
2019	M	≥College	41·94 (40·88, 43·04)	-6·15 (-7·18, -5·07)	-12·78 (-14·94, -10·54)
2019	F	≤High school	24·25 (23·58, 24·98)	-3·96 (-4·58, -3·25)	-14·04 (-16·23, -11·51)
2019	F	Some college	24·13 (23·52, 24·82)	-3·94 (-4·54, -3·26)	-14·05 (-16·19, -11·60)

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
2019	F	≥College	23·82 (23·22, 24·64)	-3·68 (-4·29, -2·91)	-13·39 (-15·60, -10·56)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	43·46 (42·13, 44·89)	-5·31 (-6·66, -3·89)	-10·88 (-13·64, -7·97)
2019	M	Some college	43·33 (42·21, 44·70)	-5·34 (-6·53, -4·06)	-10·97 (-13·40, -8·33)
2019	M	≥College	43·03 (42·04, 44·08)	-5·06 (-6·02, -4·02)	-10·52 (-12·54, -8·35)
2019	F	≤High school	25·03 (24·40, 25·73)	-3·18 (-3·80, -2·50)	-11·27 (-13·46, -8·84)
2019	F	Some college	25·08 (24·53, 25·71)	-3·00 (-3·54, -2·39)	-10·68 (-12·60, -8·52)
2019	F	≥College	25·06 (24·63, 25·51)	-2·45 (-2·87, -2·00)	-8·90 (-10·45, -7·29)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	40·06 (37·83, 42·43)	-8·71 (-10·96, -6·35)	-17·86 (-22·46, -13·02)
2019	M	Some college	40·00 (38·09, 42·21)	-8·67 (-10·66, -6·55)	-17·81 (-21·86, -13·43)
2019	M	≥College	40·02 (38·40, 41·75)	-8·07 (-9·66, -6·34)	-16·78 (-20·09, -13·19)
2019	F	≤High school	23·17 (22·15, 24·32)	-5·04 (-6·07, -3·91)	-17·86 (-21·51, -13·85)
2019	F	Some college	23·40 (22·51, 24·42)	-4·68 (-5·58, -3·68)	-16·68 (-19·85, -13·11)
2019	F	≥College	23·84 (23·15, 24·57)	-3·67 (-4·35, -2·94)	-13·33 (-15·82, -10·70)
Alcohol use category 3					
Reference scenario (no policy)					
2000	M	≤High school	104·15 (103·87, 104·41)	·	·
2000	M	Some college	99·42 (99·15, 99·60)	·	·
2000	M	≥College	93·35 (93·03, 93·63)	·	·
2000	F	≤High school	71·89 (71·61, 72·20)	·	·
2000	F	Some college	67·50 (67·11, 67·83)	·	·
2000	F	≥College	61·34 (61·11, 61·54)	·	·
2019	M	≤High school	101·84 (101·65, 102·11)	·	·
2019	M	Some college	96·37 (96·07, 96·64)	·	·
2019	M	≥College	91·00 (90·65, 91·29)	·	·
2019	F	≤High school	70·06 (69·72, 70·39)	·	·
2019	F	Some college	65·19 (64·85, 65·47)	·	·
2019	F	≥College	59·79 (59·55, 60·02)	·	·
Scenario 1: Price increase by 10% for beer, wine, and spirits					
2019	M	≤High school	98·79 (98·20, 99·58)	-3·05 (-3·70, -2·35)	-2·99 (-3·62, -2·30)
2019	M	Some college	93·29 (92·64, 94·06)	-3·08 (-3·65, -2·44)	-3·20 (-3·79, -2·53)
2019	M	≥College	87·94 (87·43, 88·81)	-3·06 (-3·64, -2·48)	-3·37 (-3·99, -2·71)
2019	F	≤High school	67·48 (66·81, 68·19)	-2·58 (-3·04, -2·10)	-3·69 (-4·33, -3·00)
2019	F	Some college	62·77 (62·28, 63·38)	-2·42 (-2·85, -1·99)	-3·71 (-4·37, -3·05)
2019	F	≥College	57·67 (57·16, 58·12)	-2·13 (-2·48, -1·77)	-3·56 (-4·16, -2·96)

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
Scenario 2: Price increase by 30% for beer, wine, and spirits					
2019	M	≤High school	92·69 (90·89, 94·88)	-9·15 (-11·09, -7·04)	-8·98 (-10·87, -6·91)
2019	M	Some college	87·13 (85·34, 89·17)	-9·25 (-10·96, -7·33)	-9·59 (-11·38, -7·60)
2019	M	≥College	81·81 (80·33, 83·85)	-9·19 (-10·92, -7·43)	-10·10 (-11·97, -8·14)
2019	F	≤High school	62·31 (61·00, 63·86)	-7·75 (-9·12, -6·30)	-11·06 (-13·00, -9·01)
2019	F	Some college	57·94 (56·65, 59·31)	-7·26 (-8·54, -5·96)	-11·13 (-13·10, -9·14)
2019	F	≥College	53·41 (52·19, 54·58)	-6·38 (-7·45, -5·31)	-10·67 (-12·49, -8·87)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	93·25 (91·45, 95·42)	-8·59 (-10·53, -6·51)	-8·44 (-10·33, -6·38)
2019	M	Some college	87·91 (86·13, 89·93)	-8·46 (-10·17, -6·57)	-8·78 (-10·56, -6·81)
2019	M	≥College	83·09 (81·67, 85·08)	-7·91 (-9·55, -6·20)	-8·69 (-10·47, -6·79)
2019	F	≤High school	63·43 (62·12, 64·76)	-6·64 (-7·94, -5·23)	-9·47 (-11·31, -7·47)
2019	F	Some college	59·34 (58·14, 60·67)	-5·86 (-7·05, -4·61)	-8·98 (-10·81, -7·08)
2019	F	≥College	55·10 (54·12, 56·13)	-4·70 (-5·65, -3·71)	-7·86 (-9·45, -6·20)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	87·70 (84·61, 91·26)	-14·13 (-17·37, -10·66)	-13·88 (-17·03, -10·46)
2019	M	Some college	82·54 (79·62, 85·80)	-13·84 (-16·69, -10·71)	-14·36 (-17·33, -11·09)
2019	M	≥College	78·25 (75·78, 81·36)	-12·75 (-15·46, -9·92)	-14·01 (-16·95, -10·87)
2019	F	≤High school	59·37 (57·24, 61·63)	-10·69 (-12·83, -8·36)	-15·26 (-18·29, -11·94)
2019	F	Some college	55·90 (53·94, 57·95)	-9·29 (-11·25, -7·24)	-14·26 (-17·26, -11·11)
2019	F	≥College	52·53 (50·95, 54·19)	-7·27 (-8·81, -5·65)	-12·15 (-14·74, -9·44)

g per day: grams of pure alcohol per day, CI: Credible interval, minimum and maximum of the simulated policy effect across 60 model iterations, Alcohol use category 1: up to 20 g per day for females and up to 40 g per day for males, Alcohol use category 2: more than 20 g per day up to 40 g per day for females and more than 40 g per day up to 60 g per day for males, Alcohol use category 3: more than 40 g per day for females and more than 60 g per day for males, M: Males, F: Females, ≤High school: High school degree or less, ≥College: College degree or more. Less conservative policy setting: $r=0.4$, application of beverage-nonspecific own-price participation elasticity.

Supplement 6C. Simulated average consumption levels by alcohol use category as of 2018 (sensitivity analysis 2, less conservative policy setting).

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
Alcohol use category 1					
Reference scenario (no policy)					
2018	M	≤High school	10·19 (10·17, 10·21)	·	·
2018	M	Some college	10·00 (9·97, 10·03)	·	·
2018	M	≥College	10·93 (10·91, 10·96)	·	·
2018	F	≤High school	4·10 (4·09, 4·10)	·	·
2018	F	Some college	4·40 (4·38, 4·42)	·	·
2018	F	≥College	5·09 (5·07, 5·10)	·	·
2019	M	≤High school	9·75 (9·73, 9·78)	·	·
2019	M	Some college	9·53 (9·51, 9·56)	·	·
2019	M	≥College	10·37 (10·34, 10·40)	·	·
2019	F	≤High school	3·94 (3·93, 3·95)	·	·
2019	F	Some college	4·22 (4·20, 4·23)	·	·
2019	F	≥College	4·87 (4·85, 4·88)	·	·
Scenario 1: Price increase by 10% for beer, wine, and spirits					
2019	M	≤High school	8·94 (8·81, 9·06)	-0·81 (-0·93, -0·69)	-8·33 (-9·51, -7·08)
2019	M	Some college	8·73 (8·62, 8·86)	-0·80 (-0·90, -0·69)	-8·41 (-9·45, -7·27)
2019	M	≥College	9·50 (9·42, 9·62)	-0·87 (-0·96, -0·77)	-8·41 (-9·28, -7·41)
2019	F	≤High school	3·60 (3·56, 3·64)	-0·34 (-0·37, -0·30)	-8·64 (-9·49, -7·64)
2019	F	Some college	3·85 (3·81, 3·90)	-0·36 (-0·40, -0·32)	-8·64 (-9·41, -7·67)
2019	F	≥College	4·45 (4·41, 4·50)	-0·42 (-0·46, -0·36)	-8·58 (-9·46, -7·35)
Scenario 2: Price increase by 30% for beer, wine, and spirits					
2019	M	≤High school	7·41 (7·08, 7·76)	-2·34 (-2·66, -1·99)	-23·99 (-27·32, -20·43)
2019	M	Some college	7·23 (6·94, 7·54)	-2·31 (-2·58, -2·00)	-24·19 (-27·10, -20·99)
2019	M	≥College	7·86 (7·62, 8·16)	-2·51 (-2·76, -2·21)	-24·19 (-26·59, -21·37)
2019	F	≤High school	2·96 (2·86, 3·07)	-0·98 (-1·07, -0·87)	-24·82 (-27·22, -22·06)
2019	F	Some college	3·17 (3·07, 3·29)	-1·05 (-1·14, -0·93)	-24·87 (-27·05, -22·13)
2019	F	≥College	3·67 (3·55, 3·83)	-1·20 (-1·32, -1·03)	-24·67 (-27·09, -21·19)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	7·68 (7·33, 8·03)	-2·07 (-2·40, -1·73)	-21·28 (-24·65, -17·70)
2019	M	Some college	7·55 (7·26, 7·87)	-1·99 (-2·27, -1·68)	-20·83 (-23·80, -17·62)
2019	M	≥College	8·34 (8·10, 8·63)	-2·02 (-2·28, -1·75)	-19·53 (-21·97, -16·87)
2019	F	≤High school	3·14 (3·04, 3·25)	-0·79 (-0·89, -0·69)	-20·18 (-22·63, -17·47)

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
2019	F	Some college	3.40 (3.31, 3.50)	-0.82 (-0.90, -0.72)	-19.34 (-21.48, -17.03)
2019	F	≥College	4.00 (3.92, 4.09)	-0.87 (-0.94, -0.78)	-17.79 (-19.32, -15.97)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	6.50 (5.97, 7.05)	-3.25 (-3.77, -2.70)	-33.36 (-38.70, -27.71)
2019	M	Some college	6.44 (5.99, 6.93)	-3.09 (-3.54, -2.62)	-32.46 (-37.15, -27.41)
2019	M	≥College	7.26 (6.87, 7.71)	-3.11 (-3.51, -2.67)	-29.95 (-33.85, -25.74)
2019	F	≤High school	2.72 (2.56, 2.88)	-1.22 (-1.37, -1.05)	-31.01 (-34.89, -26.76)
2019	F	Some college	2.98 (2.84, 3.13)	-1.24 (-1.38, -1.09)	-29.40 (-32.68, -25.80)
2019	F	≥College	3.58 (3.46, 3.71)	-1.29 (-1.41, -1.16)	-26.52 (-28.99, -23.73)
Alcohol use category 2					
Reference scenario (no policy)					
2000	M	≤High school	47.81 (47.77, 47.84)	.	.
2000	M	Some college	47.69 (47.63, 47.74)	.	.
2000	M	≥College	47.21 (47.17, 47.24)	.	.
2000	F	≤High school	27.16 (27.13, 27.19)	.	.
2000	F	Some college	26.89 (26.85, 26.93)	.	.
2000	F	≥College	26.67 (26.63, 26.70)	.	.
2019	M	≤High school	48.76 (48.68, 48.87)	.	.
2019	M	Some college	48.67 (48.58, 48.79)	.	.
2019	M	≥College	48.09 (48.00, 48.19)	.	.
2019	F	≤High school	28.21 (28.13, 28.30)	.	.
2019	F	Some college	28.08 (27.99, 28.17)	.	.
2019	F	≥College	27.50 (27.42, 27.57)	.	.
Scenario 1: Price increase by 10% for beer, wine, and spirits					
2019	M	≤High school	46.42 (45.89, 47.03)	-2.34 (-2.90, -1.75)	-4.80 (-5.94, -3.59)
2019	M	Some college	46.27 (45.87, 46.87)	-2.40 (-2.88, -1.89)	-4.93 (-5.90, -3.87)
2019	M	≥College	45.65 (45.22, 46.10)	-2.44 (-2.84, -2.02)	-5.07 (-5.91, -4.19)
2019	F	≤High school	26.74 (26.46, 27.01)	-1.47 (-1.70, -1.21)	-5.21 (-6.01, -4.30)
2019	F	Some college	26.61 (26.37, 26.88)	-1.47 (-1.69, -1.21)	-5.23 (-6.00, -4.33)
2019	F	≥College	26.09 (25.85, 26.44)	-1.42 (-1.64, -1.10)	-5.14 (-5.98, -4.01)
Scenario 2: Price increase by 30% for beer, wine, and spirits					
2019	M	≤High school	41.74 (40.09, 43.52)	-7.02 (-8.70, -5.26)	-14.40 (-17.83, -10.78)
2019	M	Some college	41.47 (40.12, 43.10)	-7.20 (-8.63, -5.66)	-14.80 (-17.70, -11.60)
2019	M	≥College	40.78 (39.54, 42.06)	-7.31 (-8.52, -6.06)	-15.20 (-17.73, -12.59)
2019	F	≤High school	23.80 (23.08, 24.59)	-4.41 (-5.08, -3.64)	-15.63 (-17.99, -12.89)
2019	F	Some college	23.68 (23.02, 24.42)	-4.40 (-5.05, -3.65)	-15.68 (-17.99, -12.99)

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
2019	F	≥College	23·26 (22·57, 24·24)	-4·24 (-4·94, -3·31)	-15·42 (-17·95, -12·00)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	42·20 (40·55, 43·97)	-6·56 (-8·24, -4·81)	-13·46 (-16·89, -9·86)
2019	M	Some college	42·24 (40·88, 43·85)	-6·43 (-7·87, -4·91)	-13·21 (-16·14, -10·07)
2019	M	≥College	42·08 (40·92, 43·30)	-6·01 (-7·14, -4·81)	-12·50 (-14·86, -10·00)
2019	F	≤High school	24·69 (24·02, 25·44)	-3·53 (-4·19, -2·78)	-12·50 (-14·86, -9·86)
2019	F	Some college	24·77 (24·18, 25·43)	-3·31 (-3·89, -2·67)	-11·79 (-13·85, -9·50)
2019	F	≥College	24·78 (24·32, 25·25)	-2·73 (-3·18, -2·27)	-9·92 (-11·56, -8·24)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	37·97 (35·18, 40·90)	-10·79 (-13·61, -7·88)	-22·13 (-27·89, -16·16)
2019	M	Some college	38·20 (35·89, 40·82)	-10·47 (-12·85, -7·93)	-21·51 (-26·36, -16·27)
2019	M	≥College	38·50 (36·61, 40·50)	-9·59 (-11·45, -7·62)	-19·94 (-23·82, -15·83)
2019	F	≤High school	22·63 (21·52, 23·87)	-5·58 (-6·69, -4·36)	-19·78 (-23·72, -15·43)
2019	F	Some college	22·93 (21·98, 24·01)	-5·15 (-6·10, -4·09)	-18·34 (-21·72, -14·56)
2019	F	≥College	23·47 (22·75, 24·22)	-4·04 (-4·75, -3·29)	-14·68 (-17·28, -11·96)
Alcohol use category 3					
Reference scenario (no policy)					
2000	M	≤High school	104·15 (103·87, 104·41)	·	·
2000	M	Some college	99·42 (99·15, 99·60)	·	·
2000	M	≥College	93·35 (93·03, 93·63)	·	·
2000	F	≤High school	71·89 (71·61, 72·20)	·	·
2000	F	Some college	67·50 (67·11, 67·83)	·	·
2000	F	≥College	61·34 (61·11, 61·54)	·	·
2019	M	≤High school	101·84 (101·65, 102·11)	·	·
2019	M	Some college	96·37 (96·07, 96·64)	·	·
2019	M	≥College	91·00 (90·65, 91·29)	·	·
2019	F	≤High school	70·06 (69·72, 70·39)	·	·
2019	F	Some college	65·19 (64·85, 65·47)	·	·
2019	F	≥College	59·79 (59·55, 60·02)	·	·
Scenario 1: Price increase by 10% for beer, wine, and spirits					
2019	M	≤High school	97·53 (96·71, 98·66)	-4·31 (-5·28, -3·26)	-4·23 (-5·17, -3·20)
2019	M	Some college	92·16 (91·27, 93·17)	-4·22 (-5·01, -3·33)	-4·37 (-5·21, -3·45)
2019	M	≥College	86·91 (86·27, 88·00)	-4·09 (-4·90, -3·28)	-4·50 (-5·37, -3·60)
2019	F	≤High school	66·83 (66·09, 67·64)	-3·24 (-3·82, -2·64)	-4·62 (-5·44, -3·77)
2019	F	Some college	62·16 (61·62, 62·85)	-3·03 (-3·57, -2·50)	-4·65 (-5·47, -3·83)
2019	F	≥College	57·06 (56·43, 57·64)	-2·74 (-3·21, -2·24)	-4·58 (-5·38, -3·74)

Year	Sex	Education	Average alcohol consumption in g per day (CI)	Change in average alcohol consumption in g per day (CI)	Change in average alcohol consumption in % (CI)
Scenario 2: Price increase by 30% for beer, wine, and spirits					
2019	M	≤High school	88·83 (86·03, 92·07)	-13·00 (-15·95, -9·85)	-12·77 (-15·64, -9·67)
2019	M	Some college	83·64 (81·16, 86·44)	-12·73 (-15·13, -10·06)	-13·21 (-15·71, -10·43)
2019	M	≥College	78·65 (76·47, 81·39)	-12·35 (-14·78, -9·90)	-13·57 (-16·20, -10·84)
2019	F	≤High school	60·30 (58·67, 62·14)	-9·76 (-11·49, -7·94)	-13·93 (-16·38, -11·34)
2019	F	Some college	56·05 (54·44, 57·66)	-9·14 (-10·75, -7·53)	-14·03 (-16·49, -11·55)
2019	F	≥College	51·54 (49·97, 53·11)	-8·25 (-9·67, -6·76)	-13·80 (-16·22, -11·30)
Scenario 3: Price increase by 30% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	89·46 (86·66, 92·66)	-12·38 (-15·32, -9·26)	-12·16 (-15·02, -9·09)
2019	M	Some college	84·54 (82·08, 87·31)	-11·83 (-14·23, -9·19)	-12·28 (-14·77, -9·52)
2019	M	≥College	80·28 (78·21, 82·94)	-10·72 (-13·04, -8·34)	-11·78 (-14·29, -9·14)
2019	F	≤High school	61·69 (60·07, 63·37)	-8·37 (-10·03, -6·61)	-11·95 (-14·30, -9·45)
2019	F	Some college	57·93 (56·44, 59·48)	-7·26 (-8·75, -5·72)	-11·14 (-13·43, -8·78)
2019	F	≥College	54·08 (52·92, 55·29)	-5·71 (-6·84, -4·55)	-9·56 (-11·45, -7·61)
Scenario 4: Price increase by 50% for beer and spirits, price increase by 10% for wine					
2019	M	≤High school	81·29 (76·51, 86·60)	-20·55 (-25·47, -15·32)	-20·18 (-24·98, -15·03)
2019	M	Some college	76·85 (72·78, 81·39)	-19·53 (-23·52, -15·11)	-20·26 (-24·43, -15·65)
2019	M	≥College	73·59 (69·97, 77·83)	-17·41 (-21·27, -13·45)	-19·13 (-23·31, -14·74)
2019	F	≤High school	56·51 (53·84, 59·38)	-13·55 (-16·28, -10·60)	-19·34 (-23·21, -15·15)
2019	F	Some college	53·67 (51·22, 56·20)	-11·52 (-13·97, -8·97)	-17·67 (-21·44, -13·76)
2019	F	≥College	51·09 (49·21, 53·04)	-8·71 (-10·55, -6·80)	-14·56 (-17·66, -11·37)

g per day: grams of pure alcohol per day, CI: Credible interval, minimum and maximum of the simulated policy effect across 60 model iterations, Alcohol use category 1: up to 20 g per day for females and up to 40 g per day for males, Alcohol use category 2: more than 20 g per day up to 40 g per day for females and more than 40 g per day up to 60 g per day for males, Alcohol use category 3: more than 40 g per day for females and more than 60 g per day for males, M: Males, F: Females, ≤High school: High school degree or less, ≥College: College degree or more. More conservative policy setting: $r=0·8$, no application of beverage-nonspecific own-price participation elasticity.

Supplement 7. Differences in change in simulated average alcohol consumption by alcohol use category between reference (no price change) and scenario 4: 50% increase for beer and spirits and 10% increase for wine within population subgroups.

Year	Sex	Comparison (vs reference group)	Median difference in g per day	Mean difference in g per day (95% CI)
Alcohol use category 1				
2019	M	≤High school (vs ≥College)	-0.14	-0.13 (-0.22, -0.04)
2019	M	Some college (vs ≥College)	0.01	0.02 (-0.02, 0.05)
2019	F	≤High school (vs ≥College)	0.07	0.07 (0.04, 0.10)
2019	F	Some college (vs ≥College)	0.05	0.05 (0.03, 0.07)
Alcohol use category 2				
2019	M	<High school (vs ≥College)	-0.95	-0.94 (-1.92, -0.18)
2019	M	Some college (vs ≥College)	-0.75	-0.73 (-1.30, -0.29)
2019	F	≤High school (vs ≥College)	-1.48	-1.46 (-1.85, -1.06)
2019	F	Some college (vs ≥College)	-1.07	-1.06 (-1.31, -0.79)
Alcohol use category 3				
2019	M	≤High school (vs ≥College)	-2.31	-2.33 (-3.85, -1.11)
2019	M	Some college (vs ≥College)	-1.64	-1.62 (-2.41, -0.94)
2019	F	≤High school (vs ≥College)	-4.22	-4.20 (-5.19, -3.19)
2019	F	Some college (vs ≥College)	-2.48	-2.46 (-2.97, -1.92)

g per day: grams of pure alcohol per day, 95% CI: Credible interval, 2·5th and 97·5th percentile across 60 model iterations, Alcohol use category 1: up to 20 g per day for females and up to 40 g per day for males, Alcohol use category 2: more than 20 g per day up to 40 g per day for females and more than 40 g per day up to 60 g per day for males, Alcohol use category 3: more than 40 g per day for females and more than 60 g per day for males, M: Males, F: Females, ≤High school: High school degree or less, ≥College: College degree or more.

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